



CSE3023 Web-Based Application Development

Semester II 2025/26

Lab Report 2

Group:K1

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2.4 Task 1: Data Sharing in Servlet

2.4.1

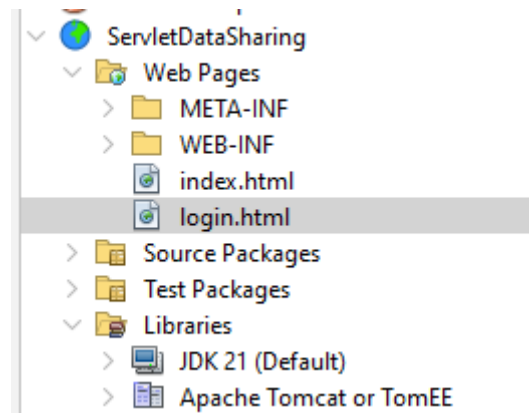


Figure 1 shows that successfully complete Step 1: Project Setup

2.4.2

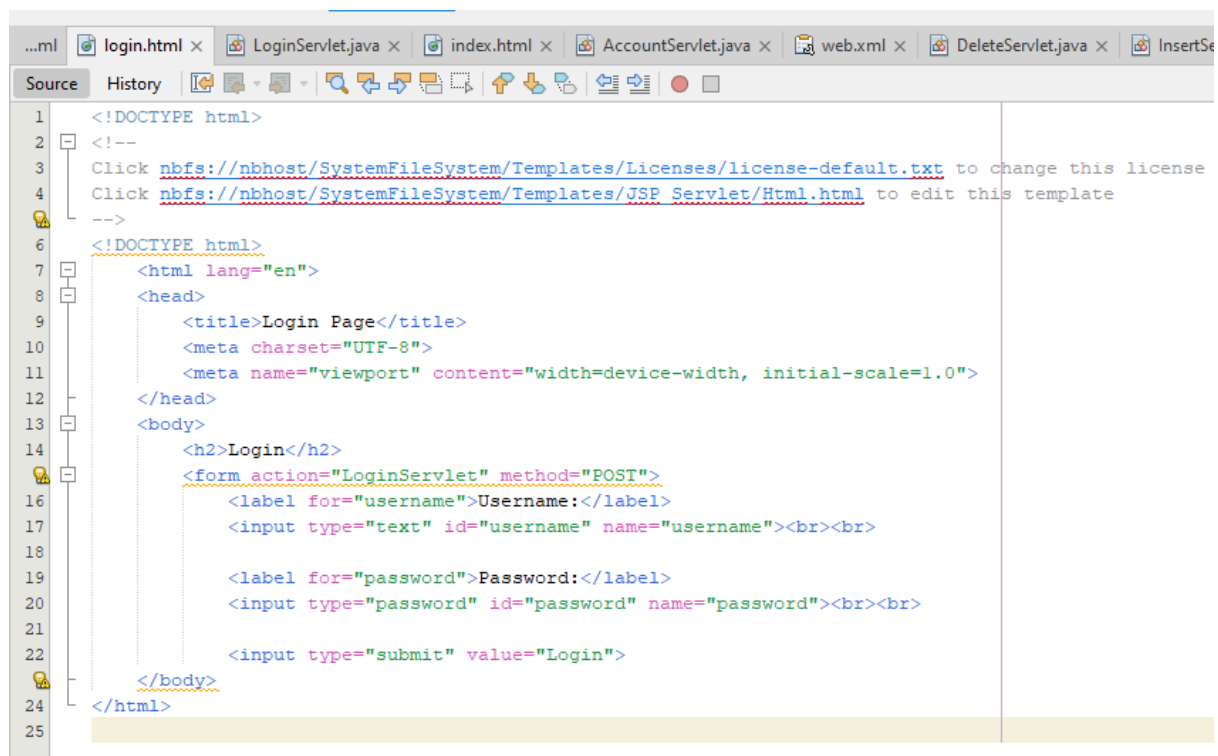
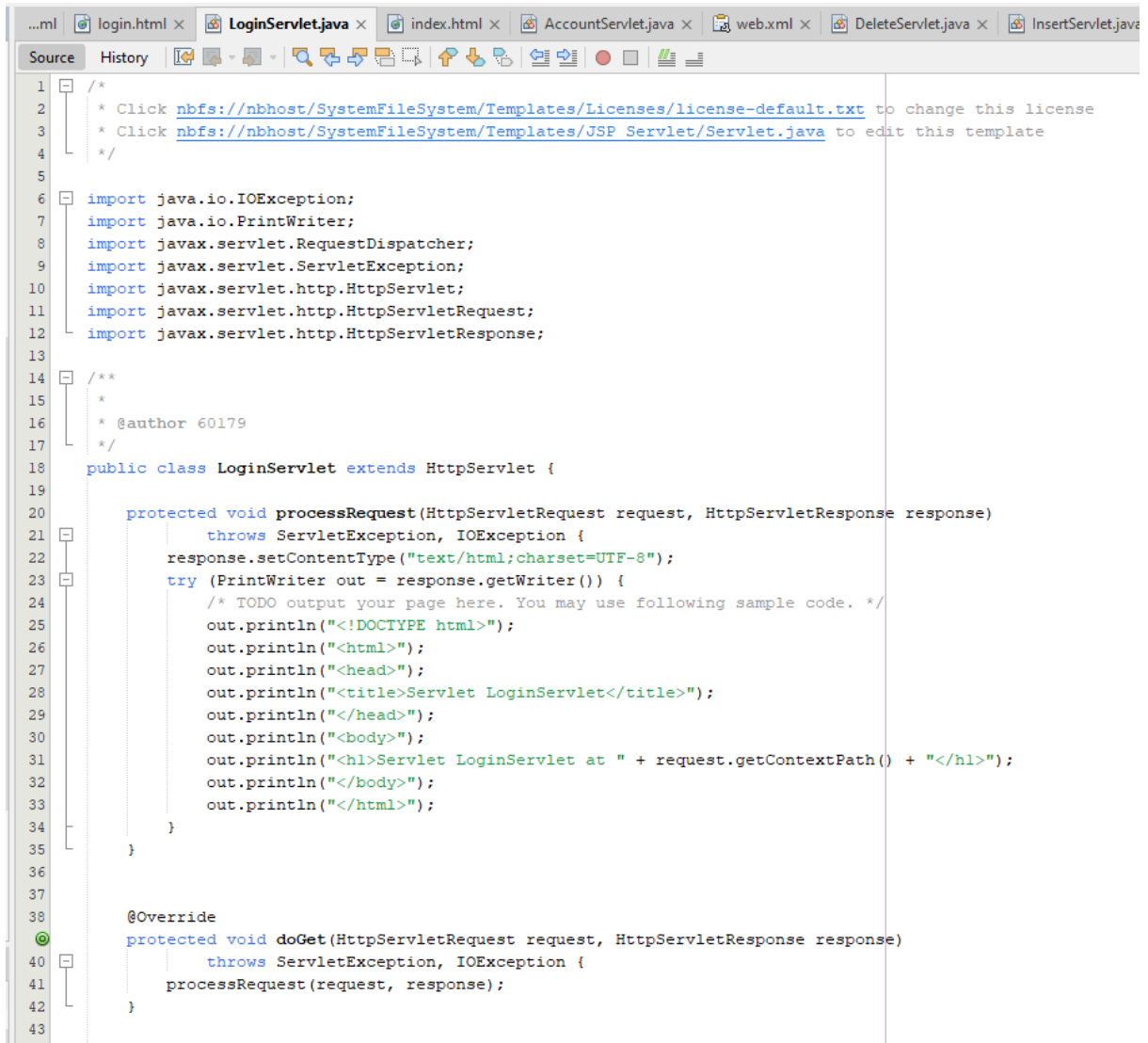


Figure 2 shows that successfully complete Step 2: Create HTML Login Page

2.4.3



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Servlet.java to edit this template
4   */
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.RequestDispatcher;
9  import javax.servlet.ServletException;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13
14 /**
15  *
16  * @author 60179
17  */
18 public class LoginServlet extends HttpServlet {
19
20     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         response.setContentType("text/html;charset=UTF-8");
23         try (PrintWriter out = response.getWriter()) {
24             /* TODO output your page here. You may use following sample code. */
25             out.println("<!DOCTYPE html>");
26             out.println("<html>");
27             out.println("<head>");
28             out.println("<title>Servlet LoginServlet</title>");
29             out.println("</head>");
30             out.println("<body>");
31             out.println("<h1>Servlet LoginServlet at " + request.getContextPath() + "</h1>");
32             out.println("</body>");
33             out.println("</html>");
34         }
35     }
36
37
38     @Override
39     protected void doGet(HttpServletRequest request, HttpServletResponse response)
40         throws ServletException, IOException {
41         processRequest(request, response);
42     }
43 }
```

Figure 3

```

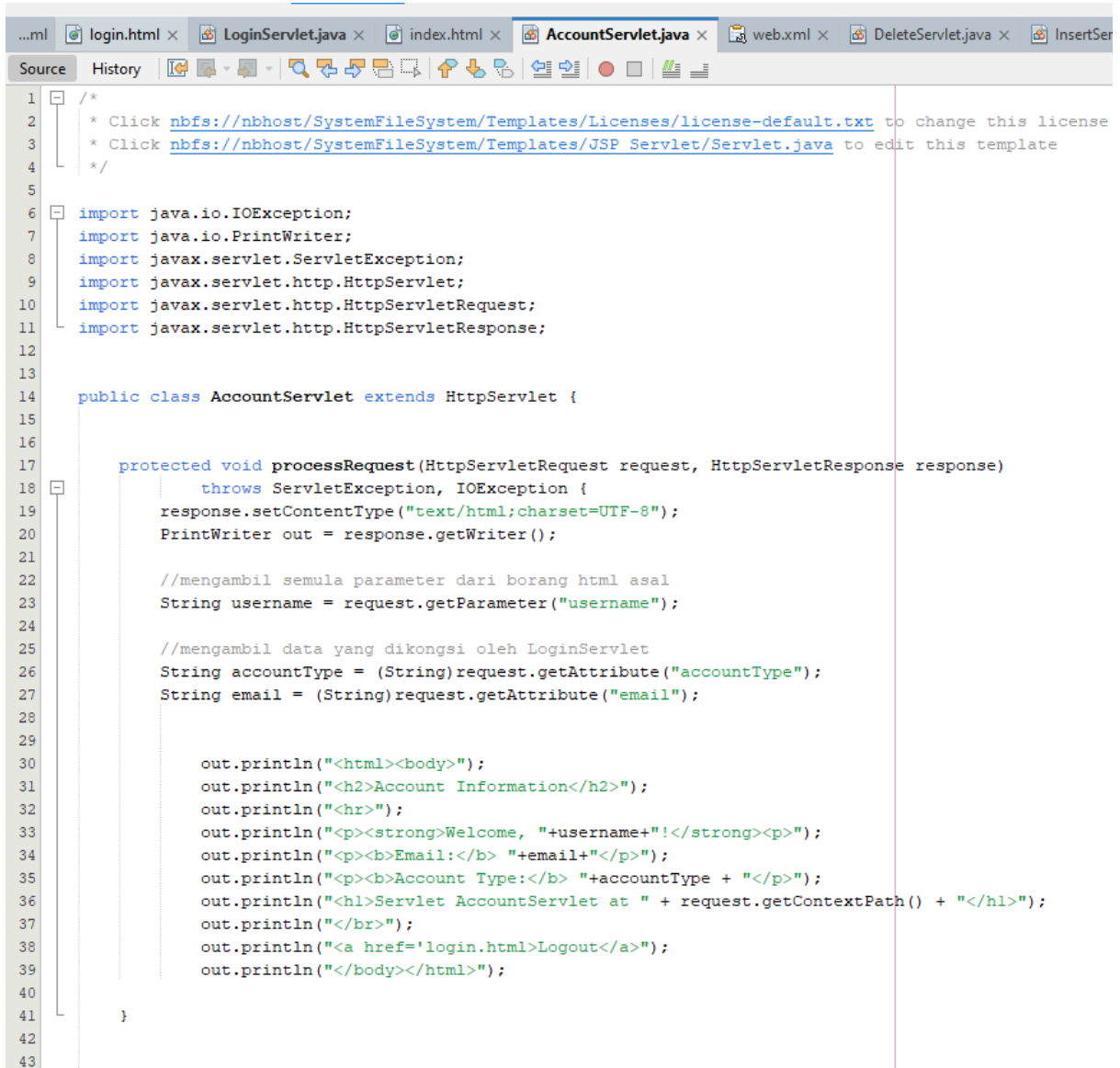
45  @Override
46  protected void doPost(HttpServletRequest request, HttpServletResponse response)
47      throws ServletException, IOException {
48
49
50      String user = request.getParameter("username");
51      String pass= request.getParameter("password");
52
53      //semakan kata laluan(authentication)
54      if(user.equals("Ahmad") && pass.equals("4567")){
55
56          //Data sharing: menyimpan maklumat tambahan ke dalam request
57          request.setAttribute("accountType", "Premium Student");
58          request.setAttribute("email", "ahmad@siswa.edu.my");
59
60          //Request Forwarding ke AccountServlet
61          RequestDispatcher rd = request.getRequestDispatcher("AccountServlet");
62          rd.forward(request,response);
63
64      }else if(user.equals("Siti") &&pass.equals("1234")){
65
66          //contoh pengguna lain
67          request.setAttribute("accountType", "Standard Student");
68          request.setAttribute("email", "siti@siswa.edu.my");
69
70          RequestDispatcher rd = request.getRequestDispatcher("AccountServlet");
71          rd.forward(request,response);
72
73      }else {
74          //if login fail
75          response.setContentType("text/html");
76          PrintWriter out =response.getWriter();
77          out.println("<h3 style = 'color:red'>Login Failed! Invalid Credentials.</h3>");
78          out.println("<a href = 'login.html'>Try Again</a>");
79      }
80
81  }
82
83  @Override
84  public String getServletInfo() {
85      return "Short description";
86  }
87  }

```

Figure 4

Figure 3 & 4 shows that successfully complete Step 3: Create a Login Servlet

2.4.4



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Servlet.java to edit this template
4   */
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.ServletException;
9  import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12
13
14 public class AccountServlet extends HttpServlet {
15
16
17     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
18         throws ServletException, IOException {
19         response.setContentType("text/html;charset=UTF-8");
20         PrintWriter out = response.getWriter();
21
22         //mengambil semula parameter dari borang html asal
23         String username = request.getParameter("username");
24
25         //mengambil data yang dikongsi oleh LoginServlet
26         String accountType = (String)request.getAttribute("accountType");
27         String email = (String)request.getAttribute("email");
28
29
30         out.println("<html><body>");
31         out.println("<h2>Account Information</h2>");
32         out.println("<hr>");
33         out.println("<p><strong>Welcome, "+username+"!</strong><p>");
34         out.println("<p><b>Email:</b> "+email+"</p>");
35         out.println("<p><b>Account Type:</b> "+accountType + "</p>");
36         out.println("<h1>Servlet AccountServlet at " + request.getContextPath() + "</h1>");
37         out.println("<br>");
38         out.println("<a href='login.html'>Logout</a>");
39         out.println("</body></html>");
40
41     }
42
43 }
```

Figure 5

```

42
43
44     @Override
45     protected void doGet(HttpServletRequest request, HttpServletResponse response)
46         throws ServletException, IOException {
47         processRequest(request, response);
48     }
49
50
51     @Override
52     protected void doPost(HttpServletRequest request, HttpServletResponse response)
53         throws ServletException, IOException {
54         processRequest(request, response);
55     }
56
57
58     @Override
59     public String getServletInfo() {
60         return "Short description";
61     } // </editor-fold>
62
63 }

```

Figure 6

Figure 5 & 6 shows that successfully complete Step 4: Create an Account Servlet

2.4.5 Output

← → ↻ ⓘ localhost:8080/ServletDataSharing/

Login

Username:

Password:

Figure 7

Account Information

Welcome, Ahmad!

Email: ahmad@siswa.edu.my

Account Type: Premium Student

Servlet AccountServlet at /ServletDataSharing

Figure 8

Figure 7 & 8 shows that successfully complete Step 5: Run the Project

Reflections

1. What have you learned from this exercise?

From this exercise, I learned how to develop a basic web application using Java Servlets. I understood how to:

- Use an HTML form to collect user input (username and password)
- Retrieve data in a servlet using `request.getParameter()`
- Perform simple user authentication using conditional statements
- Share data between servlets using `request.setAttribute()`
- Use `RequestDispatcher.forward()` to pass control from one servlet to another
- Retrieve shared data in another servlet using `request.getAttribute()`
- Display dynamic content in the browser using `PrintWriter`

Overall, this exercise helped me understand the concept of data sharing and request forwarding in servlets, which is important for building dynamic web applications.

2. What are the common methods used in Java Servlet?

Some common methods used in Java Servlet are:

doGet()

- Handles HTTP GET requests
- Used to retrieve data from the server

doPost()

- Handles HTTP POST requests
- Used to send data (e.g., form submission)

getParameter()

- Retrieves form data from the request

2.5 Task 2: Creating A Table in MySQL Database

2.5.1

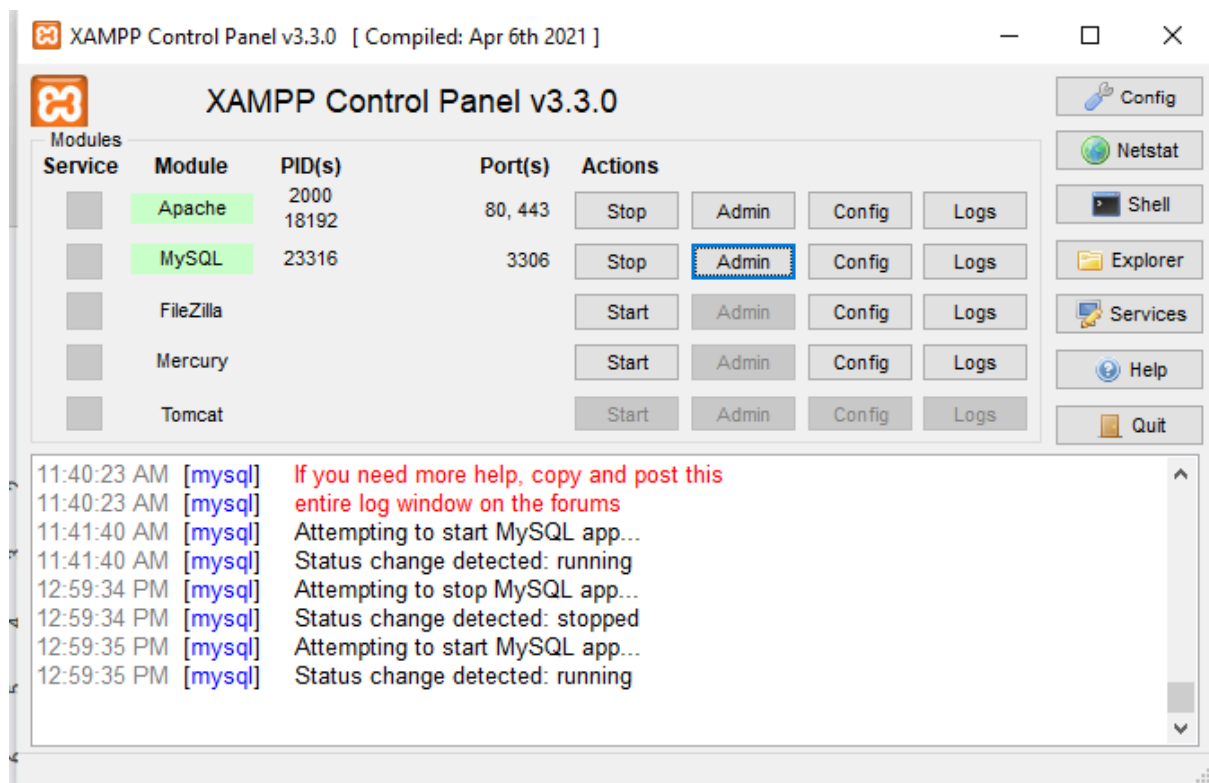


Figure 9 shows that successfully complete Step 1: Start Apache and MySQL

2.5.2

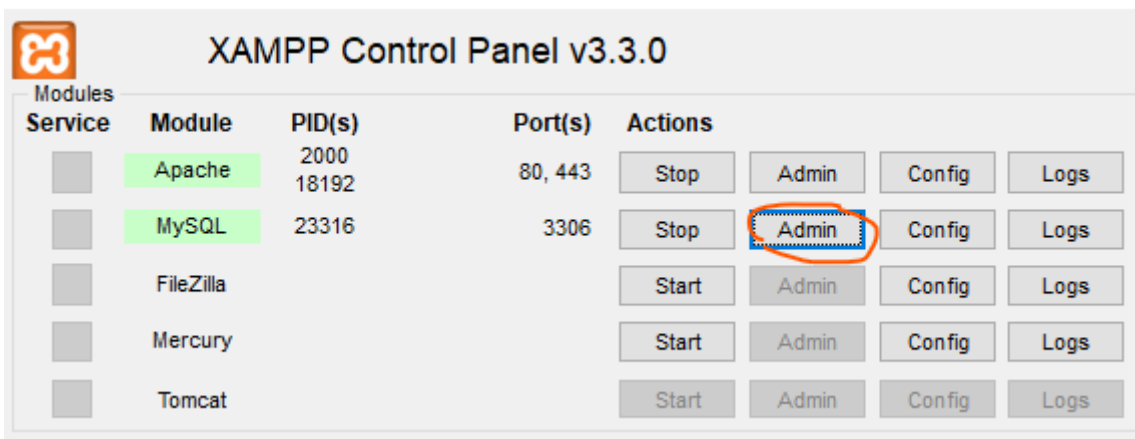


Figure 10

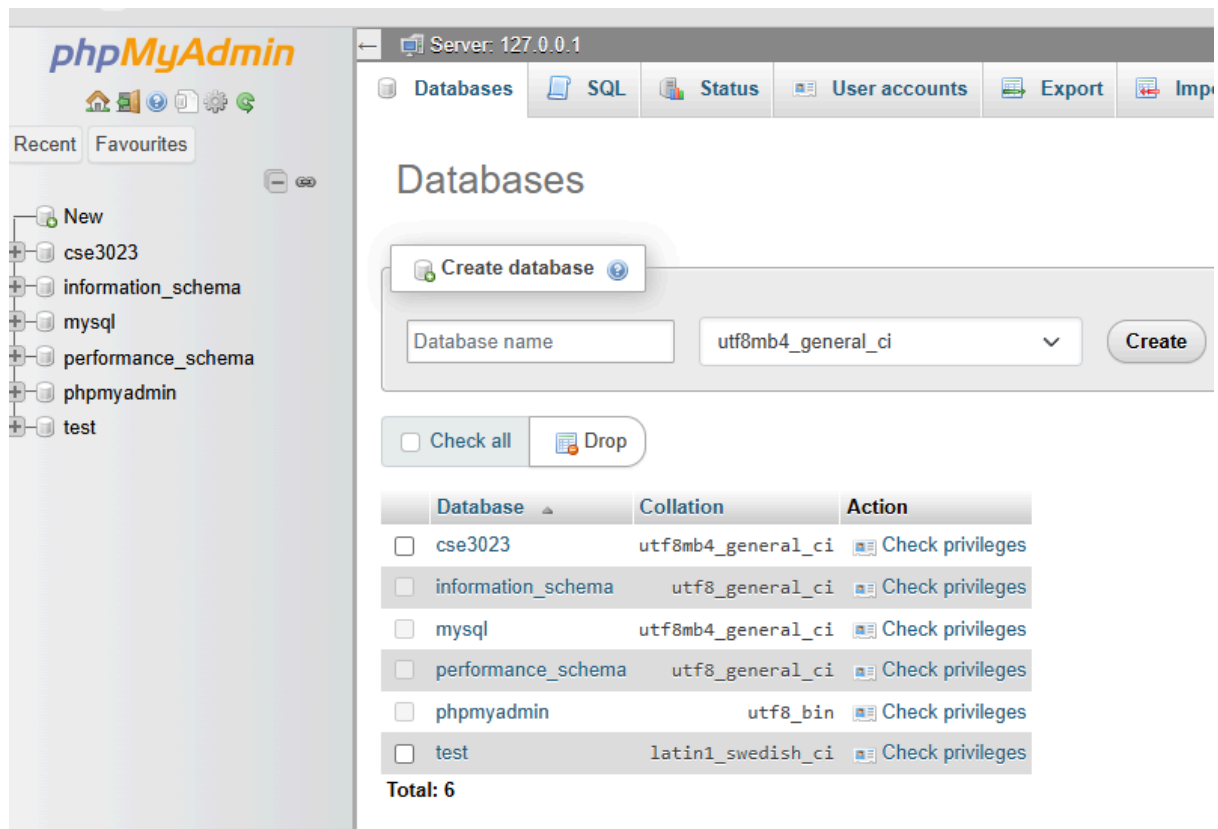
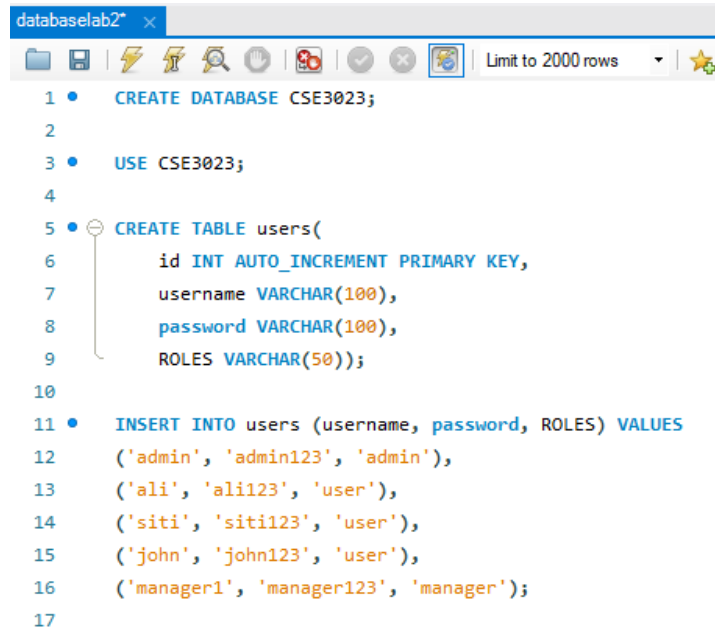


Figure 11

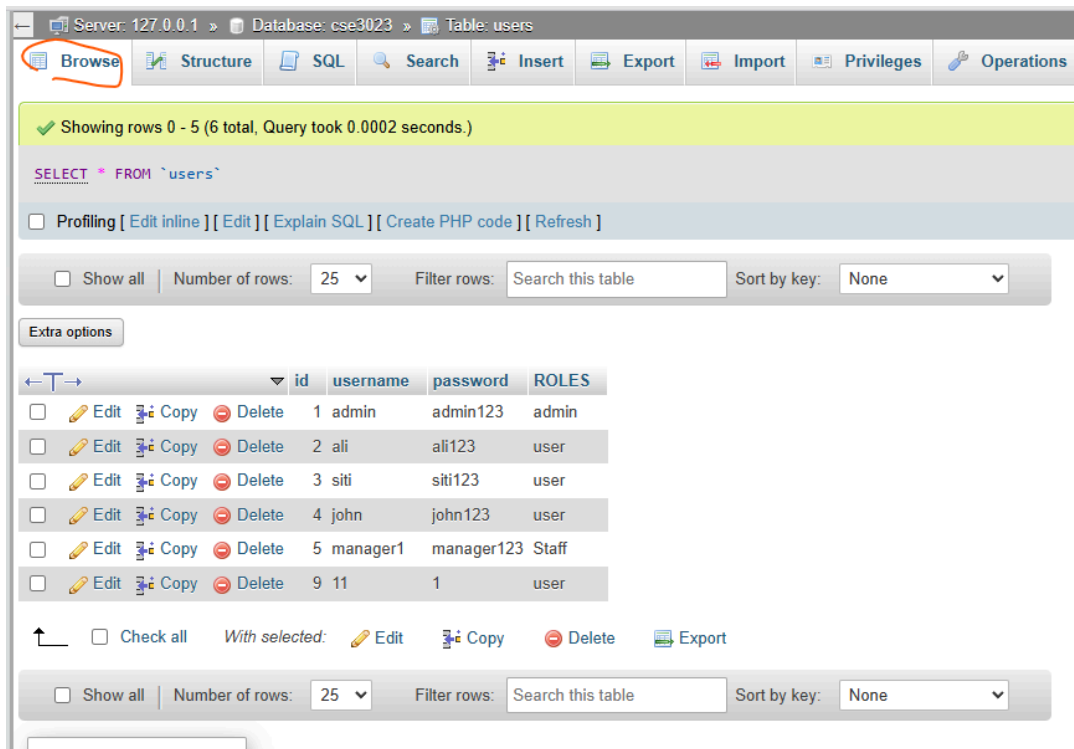
Figure 10 & 11 shows that successfully complete Step 2: Create a New Database

2.5.3 Step 3: Create a New MySQL Table



```
1 • CREATE DATABASE CSE3023;
2
3 • USE CSE3023;
4
5 • CREATE TABLE users(
6     id INT AUTO_INCREMENT PRIMARY KEY,
7     username VARCHAR(100),
8     password VARCHAR(100),
9     ROLES VARCHAR(50));
10
11 • INSERT INTO users (username, password, ROLES) VALUES
12     ('admin', 'admin123', 'admin'),
13     ('ali', 'ali123', 'user'),
14     ('siti', 'siti123', 'user'),
15     ('john', 'john123', 'user'),
16     ('manager1', 'manager123', 'manager');
17
```

Figure 12



Server: 127.0.0.1 » Database: cse3023 » Table: users

Showing rows 0 - 5 (6 total, Query took 0.0002 seconds.)

SELECT * FROM `users`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				id	username	password	ROLES
☐	Edit	Copy	Delete	1	admin	admin123	admin
☐	Edit	Copy	Delete	2	ali	ali123	user
☐	Edit	Copy	Delete	3	siti	siti123	user
☐	Edit	Copy	Delete	4	john	john123	user
☐	Edit	Copy	Delete	5	manager1	manager123	Staff
☐	Edit	Copy	Delete	9 11	1		user

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Figure 13

Figure 12 & 13 shows that successfully complete Step 3: Create a New MySQL Table

2.6 Task 3: Setting the Environment of Web Application for Database Connection

2.6.1

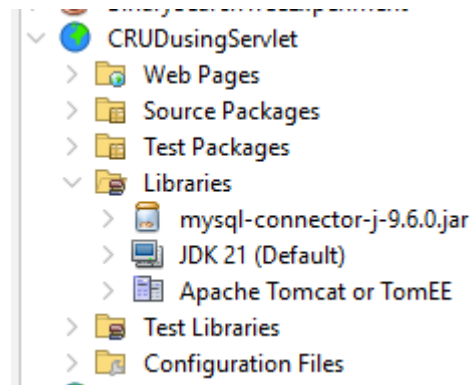


Figure 14 shows that successfully complete Step 1: Create a New Web Application Project

2.6.2

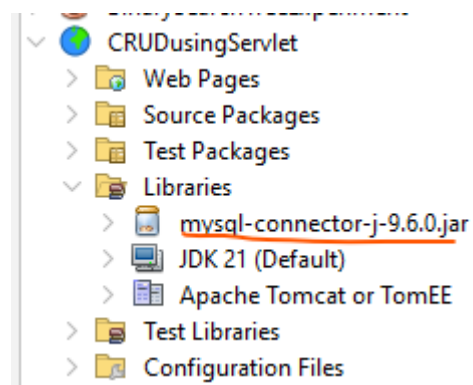
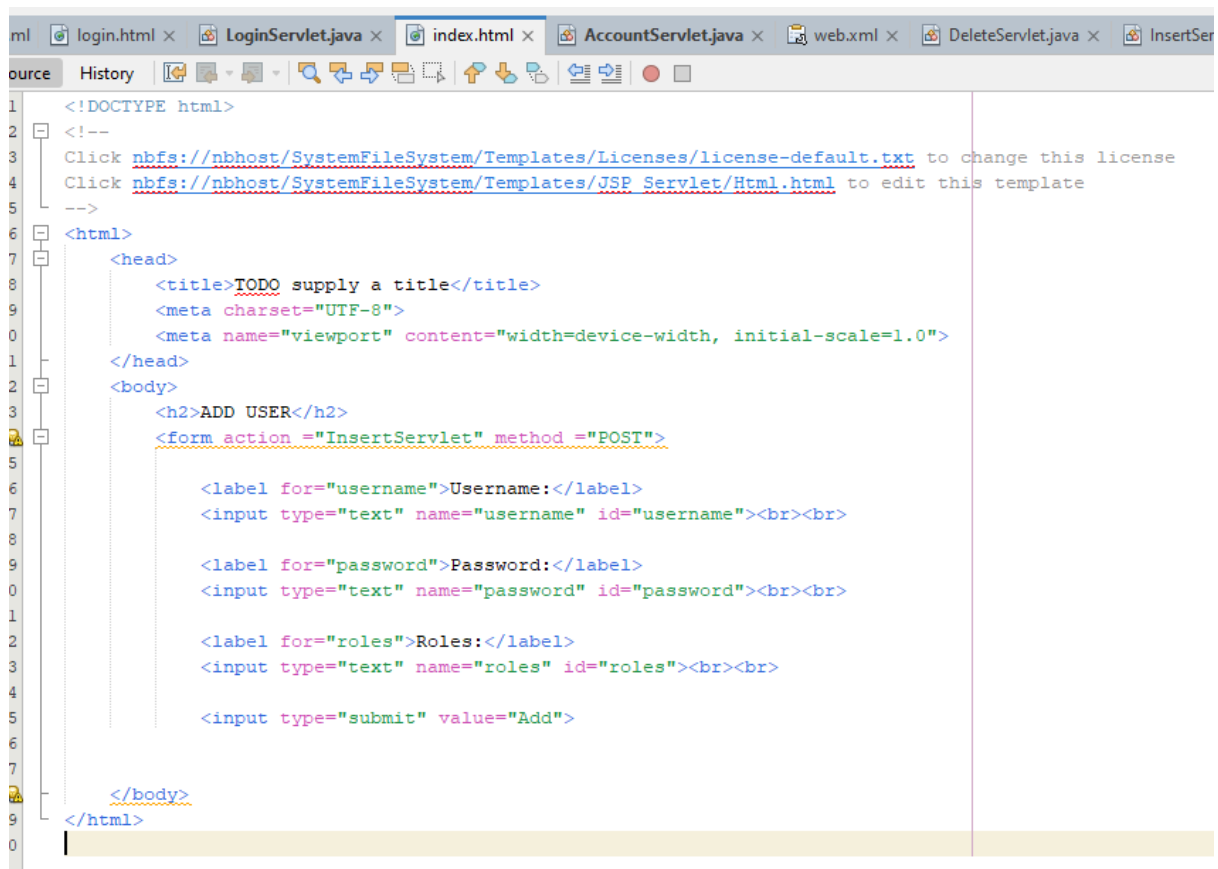


Figure 15 shows that successfully complete Step 2: Prepare the MySQL Connector

2.7 Task 4: Using Servlets for Database CRUD Operations

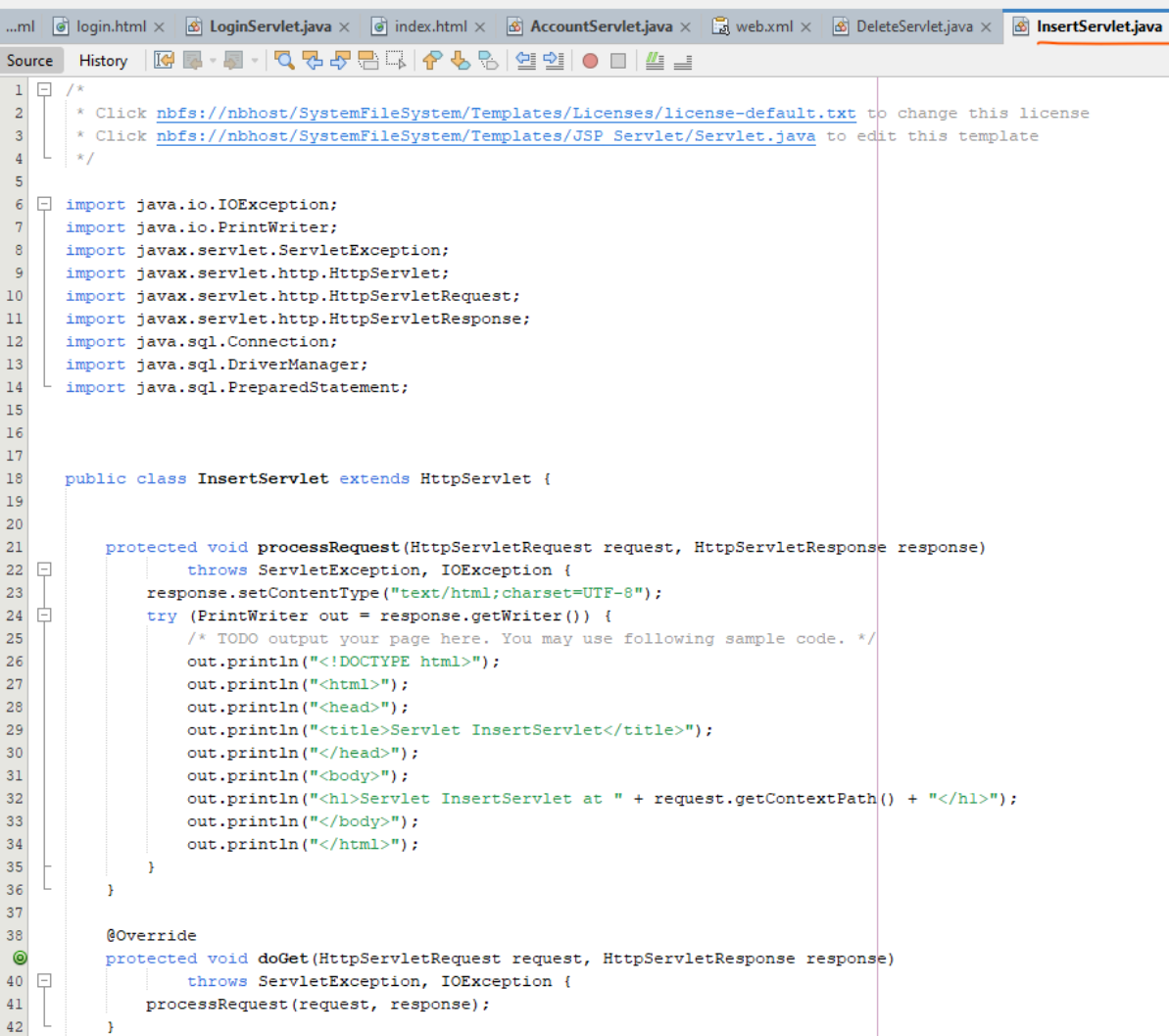
2.7.1



```
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5 -->
6 <html>
7 <head>
8 <title>TODO supply a title</title>
9 <meta charset="UTF-8">
10 <meta name="viewport" content="width=device-width, initial-scale=1.0">
11 </head>
12 <body>
13 <h2>ADD USER</h2>
14 <form action="InsertServlet" method="POST">
15
16 <label for="username">Username:</label>
17 <input type="text" name="username" id="username"><br><br>
18
19 <label for="password">Password:</label>
20 <input type="text" name="password" id="password"><br><br>
21
22 <label for="roles">Roles:</label>
23 <input type="text" name="roles" id="roles"><br><br>
24
25 <input type="submit" value="Add">
26
27 </body>
28 </html>
```

Figure 16 shows that successfully complete Step 1: Create HTML Form for Data Entry

2.7.2



```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Servlet.java to edit this template
4  */
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.ServletException;
9  import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12 import java.sql.Connection;
13 import java.sql.DriverManager;
14 import java.sql.PreparedStatement;
15
16
17
18 public class InsertServlet extends HttpServlet {
19
20
21     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
22         throws ServletException, IOException {
23         response.setContentType("text/html;charset=UTF-8");
24         try (PrintWriter out = response.getWriter()) {
25             /* TODO output your page here. You may use following sample code. */
26             out.println("<!DOCTYPE html>");
27             out.println("<html>");
28             out.println("<head>");
29             out.println("<title>Servlet InsertServlet</title>");
30             out.println("</head>");
31             out.println("<body>");
32             out.println("<h1>Servlet InsertServlet at " + request.getContextPath() + "</h1>");
33             out.println("</body>");
34             out.println("</html>");
35         }
36     }
37
38     @Override
39     protected void doGet(HttpServletRequest request, HttpServletResponse response)
40         throws ServletException, IOException {
41         processRequest(request, response);
42     }
43 }
```

Figure 17

```

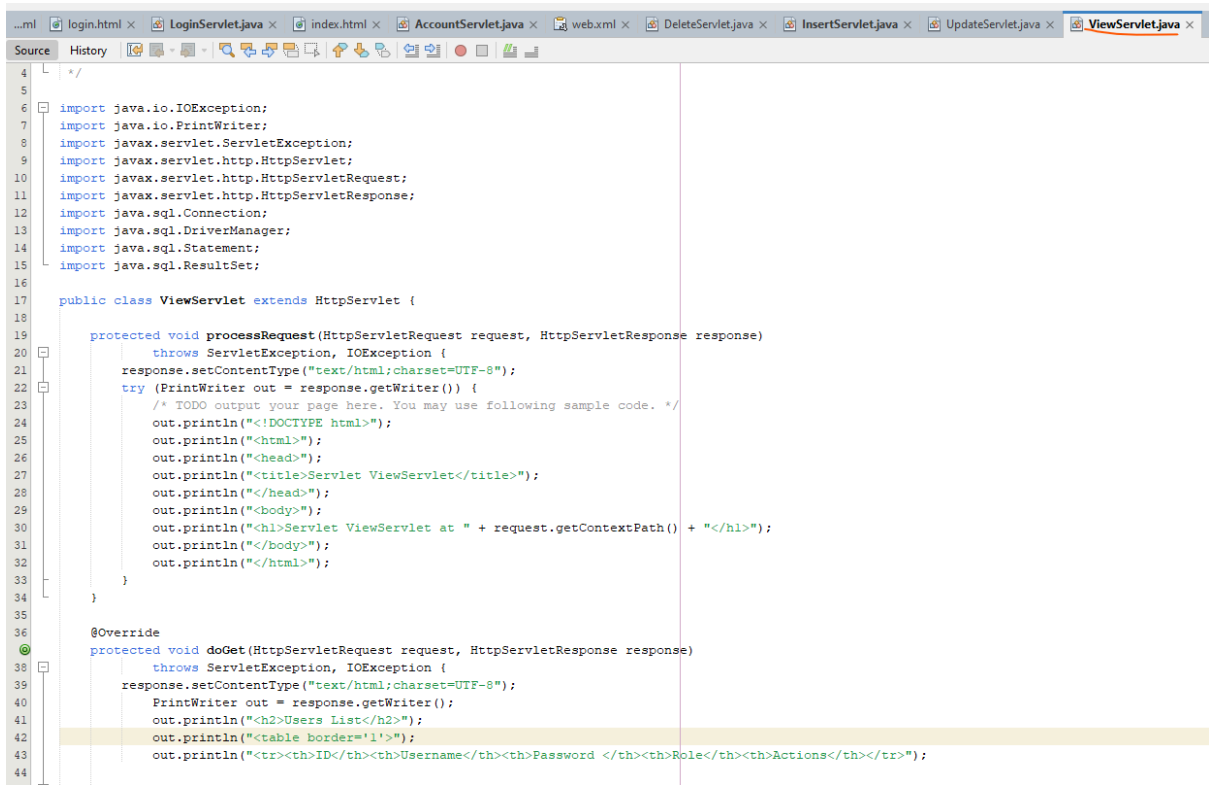
44      @Override
45      protected void doPost(HttpServletRequest request, HttpServletResponse response)
46          throws ServletException, IOException {
47          response.setContentType("text/html");
48          PrintWriter out = response.getWriter();
49
50          String username = request.getParameter("username");
51          String password= request.getParameter("password");
52          String roles= request.getParameter("roles");
53
54
55          try{
56              //load MySQL JDBC drives
57              Class.forName("com.mysql.cj.jdbc.Driver");
58              Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/CSE3023","root","admin");
59
60              String sql = "INSERT INTO users(username, password, roles)VALUES(?,?,?)";
61
62              PreparedStatement pstmt = conn.prepareStatement(sql);
63              pstmt.setString(1,username);
64              pstmt.setString(2,password);
65              pstmt.setString(3,roles);
66              pstmt.executeUpdate();
67
68              pstmt.close();
69              conn.close();
70
71              response.sendRedirect("ViewServlet");
72          }catch(Exception e){
73              e.printStackTrace();
74          }
75      }
76
77
78      @Override
79      public String getServletInfo() {
80          return "Short description";
81      } // </editor-fold>
82
83  }

```

Figure 18

Figure 17&18 shows that successfully complete Step 2: Create the Insert Servlet (Create)

2.7.3 Step 3: Create the View Servlet (Read)



```
4  * /
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.ServletException;
9  import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12 import java.sql.Connection;
13 import java.sql.DriverManager;
14 import java.sql.Statement;
15 import java.sql.ResultSet;
16
17 public class ViewServlet extends HttpServlet {
18
19     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
20         throws ServletException, IOException {
21         response.setContentType("text/html;charset=UTF-8");
22         try (PrintWriter out = response.getWriter()) {
23             /* TODO output your page here. You may use following sample code. */
24             out.println("<!DOCTYPE html>");
25             out.println("<html>");
26             out.println("<head>");
27             out.println("<title>Servlet ViewServlet</title>");
28             out.println("</head>");
29             out.println("<body>");
30             out.println("<h1>Servlet ViewServlet at " + request.getContextPath() + "</h1>");
31             out.println("</body>");
32             out.println("</html>");
33         }
34     }
35
36     @Override
37     protected void doGet(HttpServletRequest request, HttpServletResponse response)
38         throws ServletException, IOException {
39         response.setContentType("text/html;charset=UTF-8");
40         PrintWriter out = response.getWriter();
41         out.println("<h2>Users List</h2>");
42         out.println("<table border='1'>");
43         out.println("<tr><th>ID</th><th>Username</th><th>Password </th><th>Role</th><th>Actions</th></tr>");
44     }
```

Figure 19

```

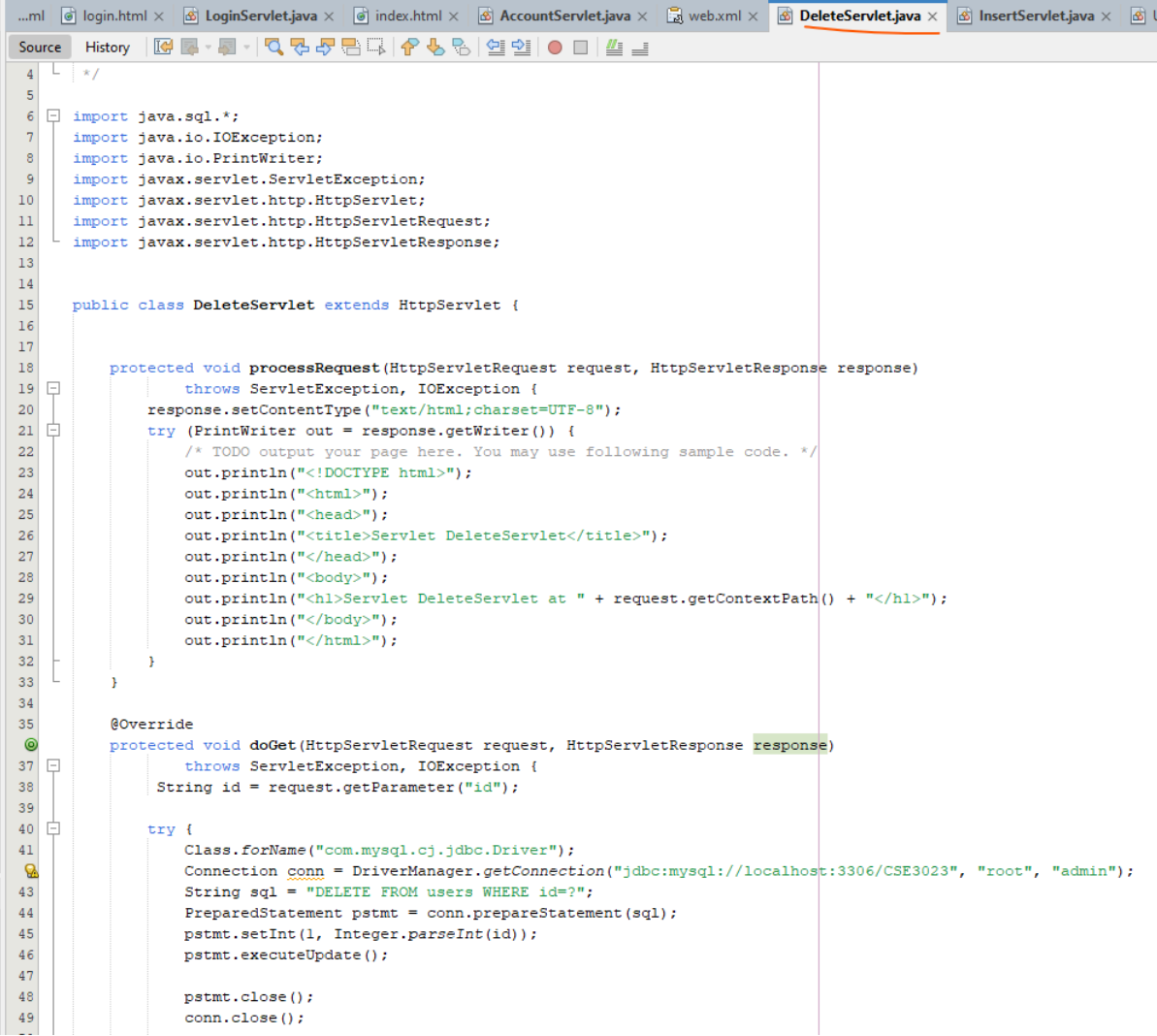
45 |
46 |         try {
47 |             //1.Load JDBC driver
48 |             Class.forName("com.mysql.cj.jdbc.Driver");
49 |             //2.Connect to database
50 |             Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/CSE3023", "root", "admin");
51 |             //3.Execute Select Query
52 |             Statement stmt = conn.createStatement();
53 |             ResultSet rs = stmt.executeQuery("SELECT * FROM users");
54 |             //4.Display result in html table
55 |             while(rs.next()){
56 |                 int id = rs.getInt("id");
57 |                 out.println("<tr>");
58 |                 out.println("<td>" + id + "</td>");
59 |                 out.println("<td>" + rs.getString("username") + "</td>");
60 |                 out.println("<td>" + rs.getString("password") + "</td>");
61 |                 out.println("<td>" + rs.getString("roles") + "</td>");
62 |
63 |                 out.println("<td><a href='UpdateServlet?id=" + id + "'>Edit</a> | ");
64 |                 out.println("<a href='DeleteServlet?id=" + id + "'>Delete</a></td>");
65 |                 out.println("</tr>");
66 |
67 |             }
68 |             out.println("</table>");
69 |             out.println("<br><a href = 'index.html'>Add New User</a>");
70 |             //5.closure
71 |             rs.close();
72 |             stmt.close();
73 |             conn.close();
74 |
75 |         } catch (Exception e) {
76 |             e.printStackTrace();
77 |             response.getWriter().println("Error: " + e.getMessage());
78 |         }
79 |
80 |     }
81 |
82 |     @Override
83 |     protected void doPost(HttpServletRequest request, HttpServletResponse response)
84 |         throws ServletException, IOException {
85 |         processRequest(request, response);
86 |     }
87 |
88 |     @Override
89 |     public String getServletInfo() {
90 |         return "Short description";
91 |     }

```

Figure 20

Figures 19 & 20 show that successfully complete Step 2: Create the Insert Servlet (Create)

2.7.4 Step 4: Create the Delete Servlet (Delete)



```
4  /*
5
6  import java.sql.*;
7  import java.io.IOException;
8  import java.io.PrintWriter;
9  import javax.servlet.ServletException;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13
14
15 public class DeleteServlet extends HttpServlet {
16
17
18     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
19         throws ServletException, IOException {
20         response.setContentType("text/html;charset=UTF-8");
21         try (PrintWriter out = response.getWriter()) {
22             /* TODO output your page here. You may use following sample code. */
23             out.println("<!DOCTYPE html>");
24             out.println("<html>");
25             out.println("<head>");
26             out.println("<title>Servlet DeleteServlet</title>");
27             out.println("</head>");
28             out.println("<body>");
29             out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
30             out.println("</body>");
31             out.println("</html>");
32         }
33     }
34
35     @Override
36     protected void doGet(HttpServletRequest request, HttpServletResponse response)
37         throws ServletException, IOException {
38         String id = request.getParameter("id");
39
40         try {
41             Class.forName("com.mysql.cj.jdbc.Driver");
42             Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/CSE3023", "root", "admin");
43             String sql = "DELETE FROM users WHERE id=?";
44             PreparedStatement pstmt = conn.prepareStatement(sql);
45             pstmt.setInt(1, Integer.parseInt(id));
46             pstmt.executeUpdate();
47
48             pstmt.close();
49             conn.close();
50         }
51     }
52 }
```

Figure 21

```

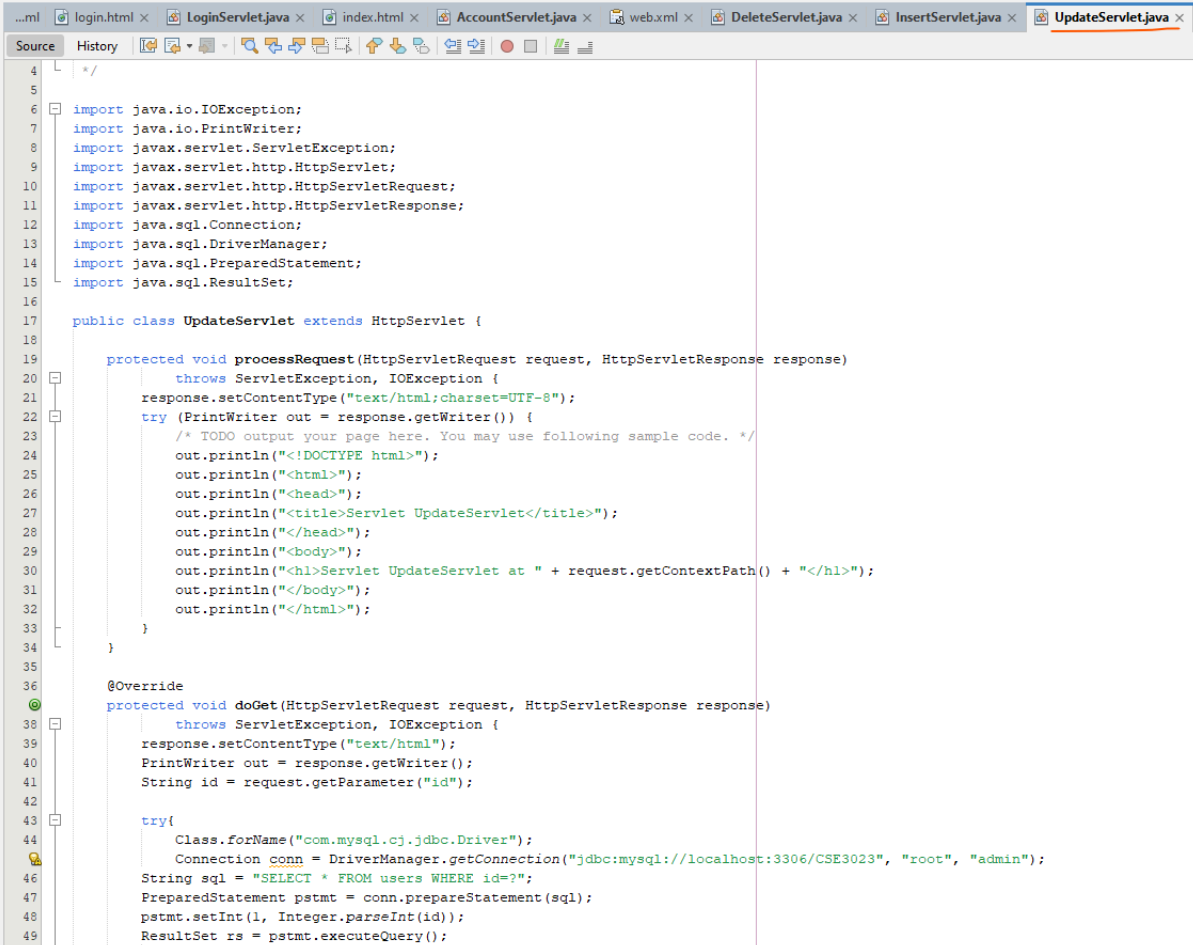
51         response.sendRedirect("ViewServlet");
52     } catch (Exception e) {
53         e.printStackTrace();
54     }
55 }
56
57 @Override
58 protected void doPost(HttpServletRequest request, HttpServletResponse response)
59     throws ServletException, IOException {
60     processRequest(request, response);
61 }
62
63 @Override
64 public String getServletInfo() {
65     return "Short description";
66 } // </editor-fold>
67
68 }
69

```

Figure 22

Figures 21 & 22 show that successfully complete Step 2: Create the Insert Servlet (Create)

2.7.5 Step 5: Create the Update Servlet (Update)



```
4  */
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.ServletException;
9  import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12 import java.sql.Connection;
13 import java.sql.DriverManager;
14 import java.sql.PreparedStatement;
15 import java.sql.ResultSet;
16
17 public class UpdateServlet extends HttpServlet {
18
19     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
20         throws ServletException, IOException {
21         response.setContentType("text/html;charset=UTF-8");
22         try (PrintWriter out = response.getWriter()) {
23             /* TODO output your page here. You may use following sample code. */
24             out.println("<!DOCTYPE html>");
25             out.println("<html>");
26             out.println("<head>");
27             out.println("<title>Servlet UpdateServlet</title>");
28             out.println("</head>");
29             out.println("<body>");
30             out.println("<h1>Servlet UpdateServlet at " + request.getContextPath() + "</h1>");
31             out.println("</body>");
32             out.println("</html>");
33         }
34     }
35
36     @Override
37     protected void doGet(HttpServletRequest request, HttpServletResponse response)
38         throws ServletException, IOException {
39         response.setContentType("text/html");
40         PrintWriter out = response.getWriter();
41         String id = request.getParameter("id");
42
43         try {
44             Class.forName("com.mysql.cj.jdbc.Driver");
45             Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/CSE3023", "root", "admin");
46             String sql = "SELECT * FROM users WHERE id=?";
47             PreparedStatement pstmt = conn.prepareStatement(sql);
48             pstmt.setInt(1, Integer.parseInt(id));
49             ResultSet rs = pstmt.executeQuery();
50         }
```

Figure 24

```

51     if (rs.next()) {
52
53         String currentRole = rs.getString("roles");
54         out.println("<h2>Update User</h2>");
55         out.println("<form action='UpdateServlet' method='POST'>");
56         out.println("<input type='hidden' name='id' value='" +
57             rs.getInt("id") + "'>");
58         out.println("Username: <input type='text' name='username' value='" + rs.getString("username") + "' required><br><br>");
59         out.println("Password: <input type='text' name='password' value='" + rs.getString("password") + "' required><br><br>");
60         out.println("Role: <select name='roles'>");
61         out.println("<option value='Admin' " + (currentRole.equals("Admin") ? "selected" : "") + ">Admin</option>");
62         out.println("<option value='Staff' " + (currentRole.equals("Staff") ? "selected" : "") + ">Staff</option>");
63         out.println("<option value='Student' " + (currentRole.equals("Student") ? "selected" : "") + ">Student</option>");
64         out.println("</select><br><br>");
65         out.println("<input type='submit' value='Update User'>");
66         out.println("</form>");
67     }
68
69     rs.close();
70     pstmt.close();
71     conn.close();
72     } catch (Exception e) {
73
74         e.printStackTrace();
75     }
76 }
77
78 @Override
79 protected void doPost(HttpServletRequest request, HttpServletResponse response)
80     throws ServletException, IOException {
81     //UpdateServlet.doPost() Method: Save Edited Record to Database
82     // Proses dan simpan data yang telah dikemaskini
83     String id = request.getParameter("id");
84     String username = request.getParameter("username");
85     String password = request.getParameter("password");
86     String roles = request.getParameter("roles");
87
88     try {
89         Class.forName("com.mysql.cj.jdbc.Driver");
90         Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/CSE3023", "root", "admin");
91         String sql = "UPDATE users SET username=?, password=?, roles=? WHERE id=?";
92         PreparedStatement pstmt = conn.prepareStatement(sql);
93         pstmt.setString(1, username);
94         pstmt.setString(2, password);
95         pstmt.setString(3, roles);
96         pstmt.setInt(4, Integer.parseInt(id));
97         pstmt.executeUpdate();
98
99         pstmt.close();
100         conn.close();
101         response.sendRedirect("ViewServlet");
102     } catch (Exception e) {
103         e.printStackTrace();
104     }
105 }
106
107 @Override
108 public String getServletInfo() {
109     return "Short description";
110 }
111 }

```

Figure 24

Figures 23 & 24 show that successfully complete Step 2: Create the Insert Servlet (Create)

Output:

[←](#) [↻](#) [localhost:8080/CRUDusingServlet/](#)

ADD USER

Username:

Password:

Roles:

Users List

ID	Username	Password	Role	Actions
1	admin	admin123	Admin	Edit Delete
2	ali	ali123	user	Edit Delete
3	siti	siti123	user	Edit Delete
4	john	john123	user	Edit Delete
5	manager1	manager123	Staff	Edit Delete
9	11	1	user	Edit Delete
12	11	1	user	Edit Delete

[Add New User](#)

Update User

Username:

Password:

Role: ▼

Users List

ID	Username	Password	Role	Actions
1	admin	admin123	Admin	Edit Delete
2	ali	ali123	user	Edit Delete
3	siti	siti123	user	Edit Delete
4	john	john123	user	Edit Delete
5	manager1	manager123	Staff	Edit Delete
9	11	1	user	Edit Delete

[Add New User](#)

2.8 Task 5: Using Servlets, DAO and Java Beans for Database CRUD Operations

2.8.1

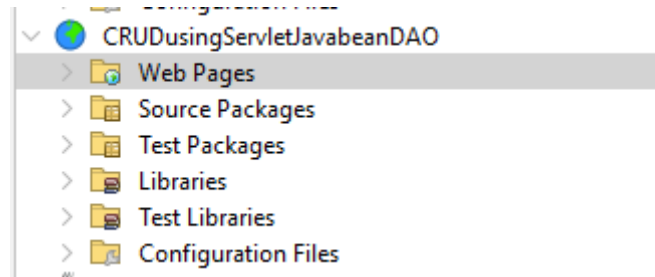


Figure 25 shows that successfully complete Step 1: Copy and Rename the Project

2.8.2 Step 2: Create the User JavaBean (Model)

```
...va User.java x UserDao.java x web.xml x DeleteServlet.java x InsertServlet.java x Up
Source History
1 package com.lab.model;
2
3 public class User {
4     private int id;
5     private String username;
6     private String password;
7     private String roles;
8     // Constructor Kosong (Wajib untuk JavaBean)
9     public User() {
10    }
11    // Constructor dengan ID (Untuk Update & Delete)
12    public User(int id, String username, String password, String roles) {
13        this.id = id;
14        this.username = username;
15        this.password = password;
16        this.roles = roles;
17    }
18    // Constructor tanpa ID (Untuk Insert)
19    public User(String username, String password, String roles) {
20        this.username = username;
21        this.password = password;
22        this.roles = roles;
23    }
24    // Getter dan Setter
25    public int getId() {
26        return id;
27    }
28
29    public void setId(int id) {
30        this.id = id;
31    }
32
33    public String getUsername() {
34        return username;
35    }
36
37    public void setUsername(String username) {
38        this.username = username;
39    }
40
41    public String getPassword() {
42        return password;
43    }
44
```

Figure 26

```

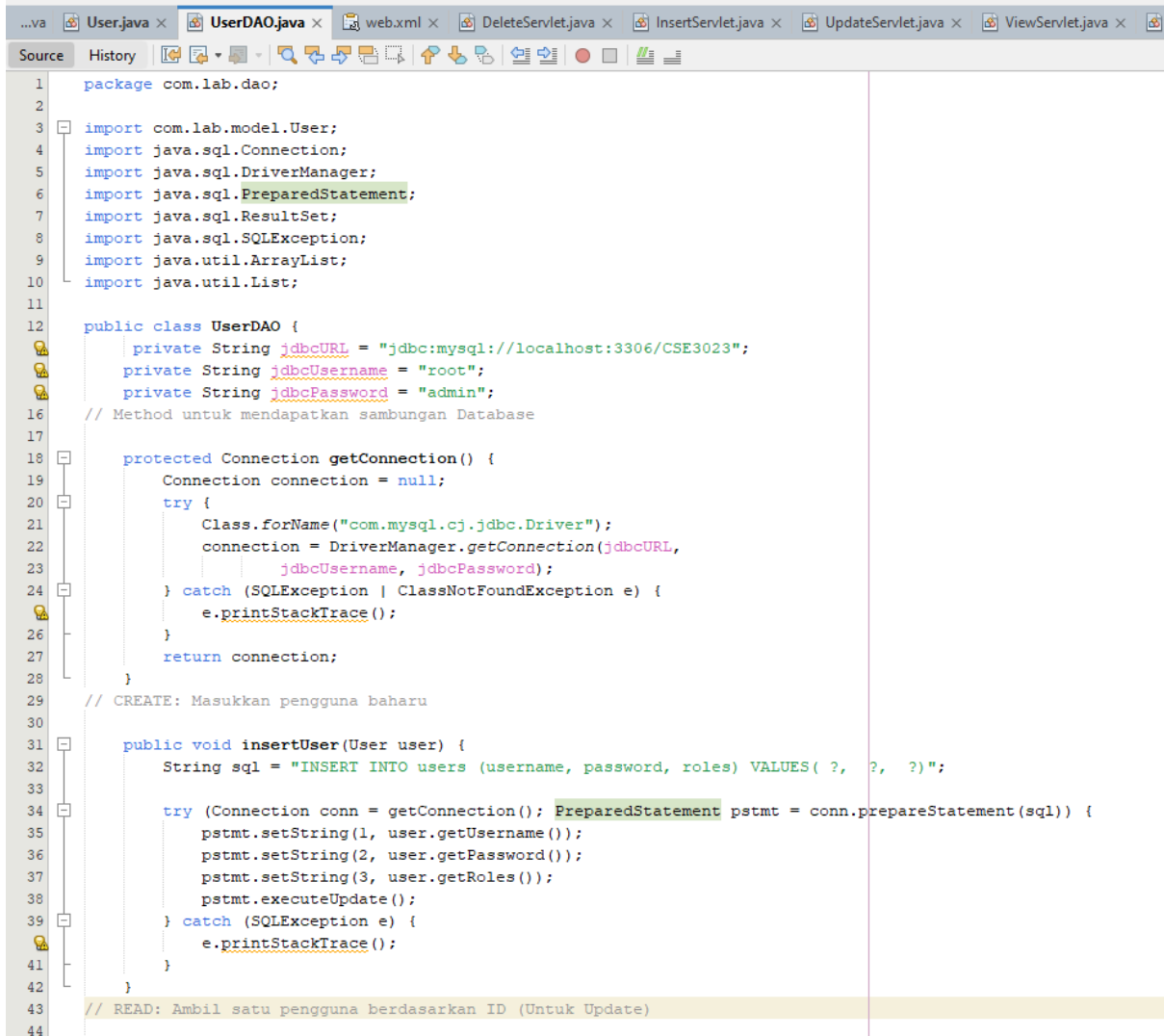
48
49     public String getRoles() {
50         return roles;
51     }
52
53     public void setRoles(String roles) {
54         this.roles = roles;
55     }
56 }
57

```

Figure 27

Figure 26 & 27 shows that successfully complete Step 1: Copy and Rename the Project

2.8.3



```

1 package com.lab.dao;
2
3 import com.lab.model.User;
4 import java.sql.Connection;
5 import java.sql.DriverManager;
6 import java.sql.PreparedStatement;
7 import java.sql.ResultSet;
8 import java.sql.SQLException;
9 import java.util.ArrayList;
10 import java.util.List;
11
12 public class UserDAO {
13     private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
14     private String jdbcUsername = "root";
15     private String jdbcPassword = "admin";
16     // Method untuk mendapatkan sambungan Database
17
18     protected Connection getConnection() {
19         Connection connection = null;
20         try {
21             Class.forName("com.mysql.cj.jdbc.Driver");
22             connection = DriverManager.getConnection(jdbcURL,
23                 jdbcUsername, jdbcPassword);
24         } catch (SQLException | ClassNotFoundException e) {
25             e.printStackTrace();
26         }
27         return connection;
28     }
29     // CREATE: Masukkan pengguna baru
30
31     public void insertUser(User user) {
32         String sql = "INSERT INTO users (username, password, roles) VALUES( ?, ?, ?)";
33
34         try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
35             pstmt.setString(1, user.getUsername());
36             pstmt.setString(2, user.getPassword());
37             pstmt.setString(3, user.getRoles());
38             pstmt.executeUpdate();
39         } catch (SQLException e) {
40             e.printStackTrace();
41         }
42     }
43     // READ: Ambil satu pengguna berdasarkan ID (Untuk Update)
44

```

Figure 28

```

43 // READ: Ambil satu pengguna berdasarkan ID (Untuk Update)
44
45 public User selectUser(int id) {
46     User user = null;
47     String sql = "SELECT id, username, password, roles FROM users WHERE id = ? ";
48
49     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
50         pstmt.setInt(1, id);
51         ResultSet rs = pstmt.executeQuery();
52         if (rs.next()) {
53             String username = rs.getString("username");
54             String password = rs.getString("password");
55             String roles = rs.getString("roles");
56             user = new User(id, username, password, roles);
57         }
58     } catch (SQLException e) {
59         e.printStackTrace();
60     }
61     return user;
62 }
63 // READ: Ambil senarai semua pengguna (Untuk View)
64
65 public List<User> selectAllUsers() {
66     List<User> users = new ArrayList<>();
67     String sql = "SELECT * FROM users";
68     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
69         ResultSet rs = pstmt.executeQuery();
70         while (rs.next()) {
71             int id = rs.getInt("id");
72             String username = rs.getString("username");
73             String password = rs.getString("password");
74             String roles = rs.getString("roles");
75             users.add(new User(id, username, password, roles));
76         }
77     } catch (SQLException e) {
78         e.printStackTrace();
79     }
80     return users;
81 }
82 // UPDATE: Kemaskini maklumat pengguna
83

```

Figure 29

```

82 // UPDATE: Kemaskini maklumat pengguna
83
84 public boolean updateUser(User user) {
85     boolean rowUpdated = false;
86     String sql = "UPDATE users SET username = ?, password = ?, roles = ? WHERE id = ?";
87
88     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
89         pstmt.setString(1, user.getUsername());
90         pstmt.setString(2, user.getPassword());
91         pstmt.setString(3, user.getRoles());
92         pstmt.setInt(4, user.getId());
93         rowUpdated = pstmt.executeUpdate() > 0;
94     } catch (SQLException e) {
95         e.printStackTrace();
96     }
97     return rowUpdated;
98 }
99 // DELETE: Padam pengguna
100
101 public boolean deleteUser(int id) {
102     boolean rowDeleted = false;
103     String sql = "DELETE FROM users WHERE id = ?";
104     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
105         pstmt.setInt(1, id);
106         rowDeleted = pstmt.executeUpdate() > 0;
107     } catch (SQLException e) {
108         e.printStackTrace();
109     }
110     return rowDeleted;
111 }
112 }

```

Figure 30

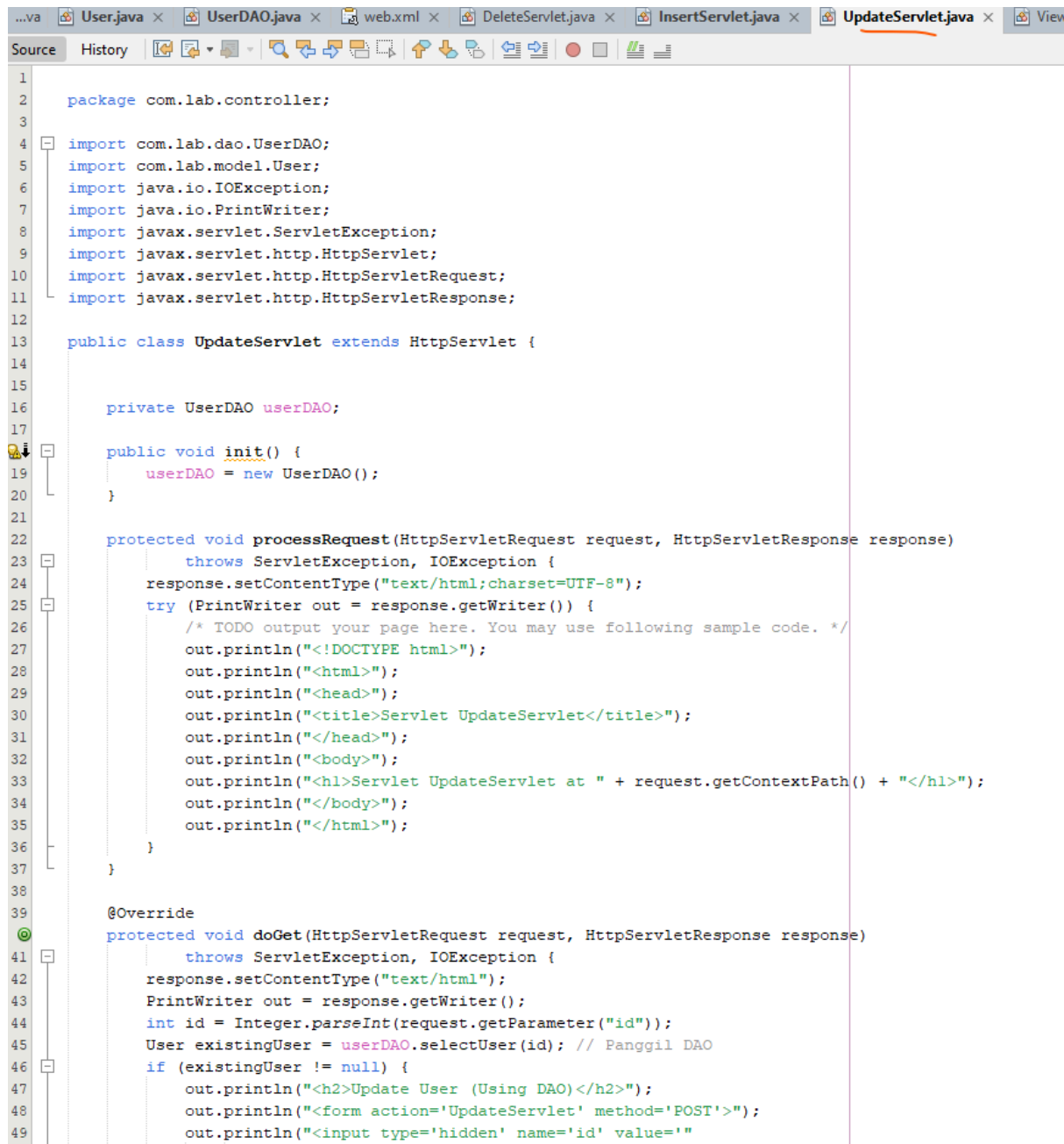
Figure 28, 29, 30 shows that successfully complete Step 3: Create the Data Access Object (UserDAO)

2.8.4 Step 4: Modify the Existing Servlets (Controller)



```
1 package com.lab.controller;
2
3 import com.lab.dao.UserDAO;
4 import com.lab.model.User;
5 import java.io.IOException;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10 import java.io.PrintWriter;
11
12
13 public class InsertServlet extends HttpServlet {
14
15     private UserDAO userDAO;
16
17     public void init() {
18         userDAO = new UserDAO();
19     }
20
21     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
22         throws ServletException, IOException {
23         response.setContentType("text/html;charset=UTF-8");
24         try (PrintWriter out = response.getWriter()) {
25             /* TODO output your page here. You may use following sample code. */
26             out.println("<!DOCTYPE html>");
27             out.println("<html>");
28             out.println("<head>");
29             out.println("<title>Servlet InsertServlet</title>");
30             out.println("</head>");
31             out.println("<body>");
32             out.println("<h1>Servlet InsertServlet at " + request.getContextPath() + "</h1>");
33             out.println("</body>");
34             out.println("</html>");
35         }
36     }
37
38     @Override
39     protected void doGet(HttpServletRequest request, HttpServletResponse response)
40         throws ServletException, IOException {
41         processRequest(request, response);
42     }
43
44     @Override
45     protected void doPost(HttpServletRequest request, HttpServletResponse response)
46         throws ServletException, IOException {
47         String username = request.getParameter("username");
48         String password = request.getParameter("password");
49         String roles = request.getParameter("roles");
50         User newUser = new User(username, password, roles);
51         userDAO.insertUser(newUser); // Panggil DAO, bukan JDBC
52         response.sendRedirect("ViewServlet");
53     }
54
55     @Override
56     public String getServletInfo() {
57         return "Short description";
58     } // </editor-fold>
59
60 }
```

Figure 31 Insert Servlet



```
1 package com.lab.controller;
2
3
4 import com.lab.dao.UserDAO;
5 import com.lab.model.User;
6 import java.io.IOException;
7 import java.io.PrintWriter;
8 import javax.servlet.ServletException;
9 import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12
13 public class UpdateServlet extends HttpServlet {
14
15     private UserDAO userDAO;
16
17     public void init() {
18         userDAO = new UserDAO();
19     }
20
21     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
22         throws ServletException, IOException {
23         response.setContentType("text/html;charset=UTF-8");
24         try (PrintWriter out = response.getWriter()) {
25             /* TODO output your page here. You may use following sample code. */
26             out.println("<!DOCTYPE html>");
27             out.println("<html>");
28             out.println("<head>");
29             out.println("<title>Servlet UpdateServlet</title>");
30             out.println("</head>");
31             out.println("<body>");
32             out.println("<h1>Servlet UpdateServlet at " + request.getContextPath() + "</h1>");
33             out.println("</body>");
34             out.println("</html>");
35         }
36     }
37
38     @Override
39     protected void doGet(HttpServletRequest request, HttpServletResponse response)
40         throws ServletException, IOException {
41         response.setContentType("text/html");
42         PrintWriter out = response.getWriter();
43         int id = Integer.parseInt(request.getParameter("id"));
44         User existingUser = userDAO.selectUser(id); // Panggil DAO
45         if (existingUser != null) {
46             out.println("<h2>Update User (Using DAO)</h2>");
47             out.println("<form action='UpdateServlet' method='POST'>");
48             out.println("<input type='hidden' name='id' value='"
```

Figure 32 Update Servlet

```

50         + existingUser.getId() + ">");
51         out.println("Username: <input type='text' name='username' value='" + existingUser.getUsername() + "' required><br><br>");
52
53         out.println("Password: <input type='text' name='password' value='" + existingUser.getPassword() + "' required><br><br>");
54
55         String currentRole = existingUser.getRoles();
56         out.println("Role: <select name='roles'>");
57         out.println("<option value='Admin' " + (currentRole.equals("Admin") ? "selected" : "") + "> Admin</option > ");
58         out.println("<option value='Staff' " + (currentRole.equals("Staff") ? "selected" : "") + "> Staff</option > ");
59         out.println("<option value='Student' " + (currentRole.equals("Student") ? "selected" : "") + "> Student</option > ");
60         out.println("</select><br><br>");
61         out.println("<input type='submit' value='Update User'>");
62         out.println("</form>");
63
64     }
65 }
66
67 @Override
68 protected void doPost(HttpServletRequest request, HttpServletResponse response)
69     throws ServletException, IOException {
70     int id = Integer.parseInt(request.getParameter("id"));
71     String username = request.getParameter("username");
72     String password = request.getParameter("password");
73     String roles = request.getParameter("roles");
74     User user = new User(id, username, password, roles);
75     userDao.updateUser(user); // Panggil DAO
76     response.sendRedirect("ViewServlet");
77
78 }
79
80 @Override
81 public String getServletInfo() {
82     return "Short description";
83 } // </editor-fold>
84
85 }

```

Figure 33 Update Servlet

```

...va User.java x UserDao.java x web.xml x DeleteServlet.java x InsertServlet.java x UpdateServlet.java x
Source History
1 package com.lab.controller;
2
3 import com.lab.dao.UserDAO;
4 import com.lab.model.User;
5 import java.io.PrintWriter;
6 import java.io.IOException;
7 import javax.servlet.ServletException;
8 import javax.servlet.http.HttpServlet;
9 import javax.servlet.http.HttpServletRequest;
10 import javax.servlet.http.HttpServletResponse;
11
12 public class DeleteServlet extends HttpServlet {
13
14     private UserDAO userDAO;
15
16     public void init() {
17         userDAO = new UserDAO();
18     }
19
20     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         response.setContentType("text/html;charset=UTF-8");
23         try (PrintWriter out = response.getWriter()) {
24             /* TODO output your page here. You may use following sample code. */
25             out.println("<!DOCTYPE html>");
26             out.println("<html>");
27             out.println("<head>");
28             out.println("<title>Servlet DeleteServlet</title>");
29             out.println("</head>");
30             out.println("<body>");
31             out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
32             out.println("</body>");
33             out.println("</html>");
34         }
35     }
36
37     @Override
38     protected void doGet(HttpServletRequest request, HttpServletResponse response)
39         throws ServletException, IOException {
40         int id = Integer.parseInt(request.getParameter("id"));
41         userDAO.deleteUser(id); // Panggil DAO
42         response.sendRedirect("ViewServlet");
43     }
44
45     @Override
46     protected void doPost(HttpServletRequest request, HttpServletResponse response)
47         throws ServletException, IOException {
48         processRequest(request, response);
49     }

```

```

51
52
53 @Override
54 public String getServletInfo() {
55     return "Short description";
56 } // </editor-fold>
57

```

Figure 34 DeleteServlet

```

...va User.java x UserDao.java x web.xml x DeleteServlet.java x InsertServlet.java x UpdateServlet.java x ViewServlet.java x Product.java x Prod
Source History
1 package com.lab.controller;
2
3 import java.io.IOException;
4 import java.io.PrintWriter;
5 import javax.servlet.ServletException;
6 import javax.servlet.http.HttpServlet;
7 import javax.servlet.http.HttpServletRequest;
8 import javax.servlet.http.HttpServletResponse;
9 import com.lab.dao.UserDAO;
10 import com.lab.model.User;
11 import java.util.List;
12
13 public class ViewServlet extends HttpServlet {
14     private UserDAO userDAO;
15
16     public void init() {
17         userDAO = new UserDAO();
18     }
19
20     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         response.setContentType("text/html;charset=UTF-8");
23         try (PrintWriter out = response.getWriter()) {
24             /* TODO output your page here. You may use following sample code. */
25             out.println("<!DOCTYPE html>");
26             out.println("<html>");
27             out.println("<head>");
28             out.println("<title>Servlet ViewServlet</title>");
29             out.println("</head>");
30             out.println("<body>");
31             out.println("<h1>Servlet ViewServlet at " + request.getContextPath() + "</h1>");
32             out.println("</body>");
33             out.println("</html>");
34         }
35     }
36
37     @Override
38     protected void doGet(HttpServletRequest request, HttpServletResponse response)
39         throws ServletException, IOException {
40         response.setContentType("text/html");
41         PrintWriter out = response.getWriter();
42
43         List<User> listUser = userDAO.selectAllUsers(); // Panggil DAO
44
45         out.println("<h2>User List</h2>");
46         out.println("<table border='1'><tr><th>ID</th><th>Username</th><th>Password</th><th>Role</th><th>Actions</th></tr>");
47         for (User user : listUser) {
48
49             out.println("<tr>");
50             out.println("<td>" + user.getId() + "</td>");
51             out.println("<td>" + user.getUsername() + "</td>");
52             out.println("<td>" + user.getPassword() + "</td>");
53             out.println("<td>" + user.getRoles() + "</td>");
54             out.println("<td><a href='UpdateServlet?id=" + user.getId() + "'>Edit</a> | ");
55             out.println("<a href='DeleteServlet?id=" + user.getId() + "'>Delete</a></td>");
56             out.println("</tr>");
57
58         }
59
60         out.println("</table>");
61         out.println("<br><a href='index.html'>Add New User</a>");
62     }
63
64     @Override
65     protected void doPost(HttpServletRequest request, HttpServletResponse response)
66         throws ServletException, IOException {
67         processRequest(request, response);
68     }
69
70     @Override
71     public String getServletInfo() {
72         return "Short description";
73     } // </editor-fold>
74
75 }

```

Figure 35 View Servlet

Output:

← ↻ ⓘ localhost:8080/CRUDusingServletJavabeanDAO/

ADD USER

Username:

Password:

Roles:

[Go to Add Product Page](#)

Before update and delete:

User List

< / tr >

ID	Username	Password	Role	Actions
1	admin	admin123	admin	Edit Delete
2	ali	ali123	user	Edit Delete
3	siti	siti123	user	Edit Delete
4	john	john123	user	Edit Delete
5	manager1	manager123	Staff	Edit Delete
9	11	1	user	Edit Delete
11	11	1	user	Edit Delete

[Add New User](#)

Update User (Using DAO)

Username:

Password:

Role: ▼

After delete:

User List

< / tr >

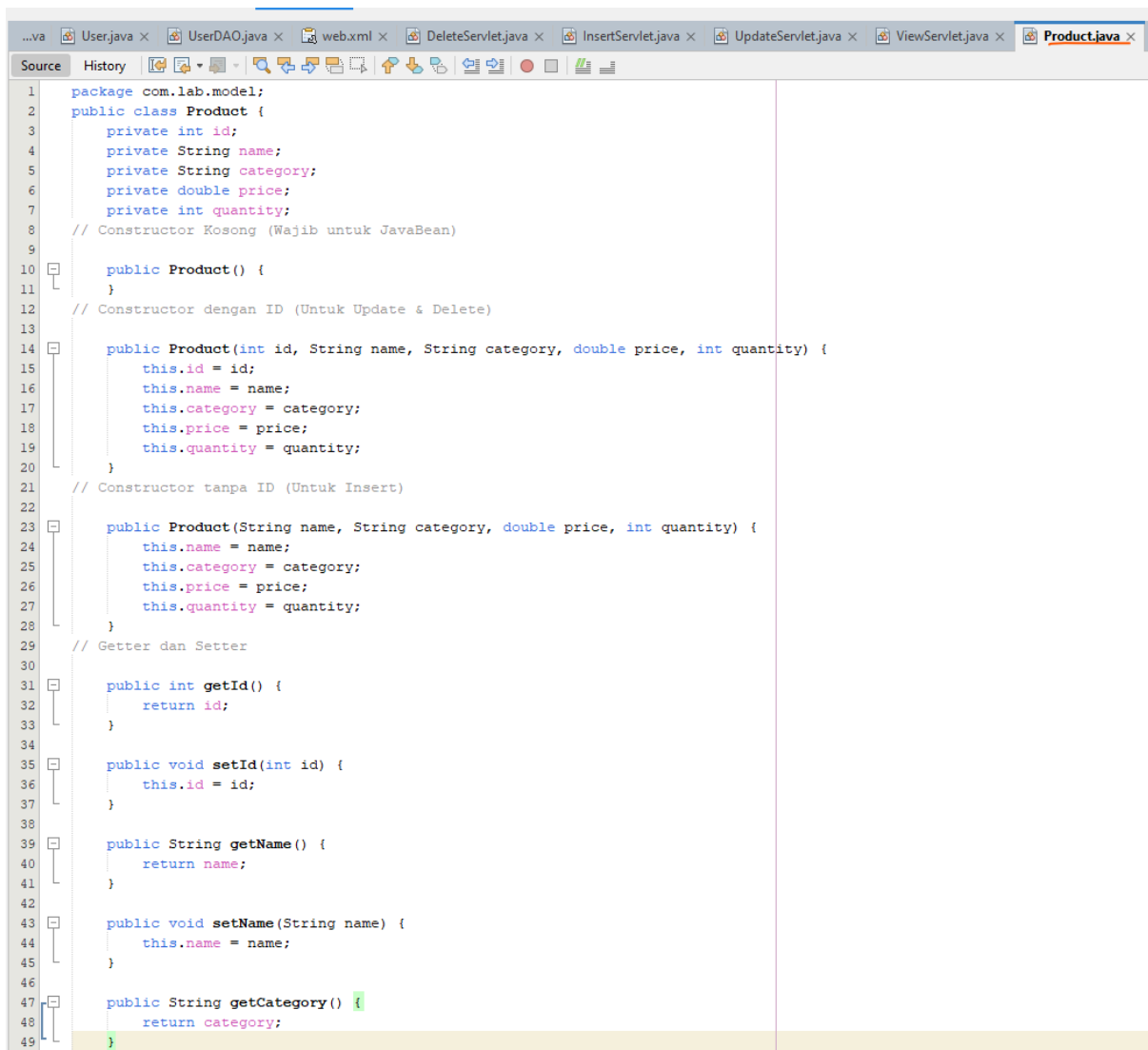
ID	Username	Password	Role	Actions
1	admin	admin123	Admin	Edit Delete
2	ali	ali123	user	Edit Delete
3	siti	siti123	user	Edit Delete
4	john	john123	user	Edit Delete
5	manager1	manager123	Staff	Edit Delete
9	11	1	user	Edit Delete

[Add New User](#)

2.9 Lab Exercise

```
18 • CREATE TABLE products (  
19     id INT AUTO_INCREMENT PRIMARY KEY,  
20     name VARCHAR(100),  
21     category VARCHAR(100),  
22     price DOUBLE,  
23     quantity INT  
24 );  
25
```

Figure 36



```
1 package com.lab.model;  
2 public class Product {  
3     private int id;  
4     private String name;  
5     private String category;  
6     private double price;  
7     private int quantity;  
8     // Constructor Kosong (Wajib untuk JavaBean)  
9  
10    public Product() {  
11    }  
12    // Constructor dengan ID (Untuk Update & Delete)  
13  
14    public Product(int id, String name, String category, double price, int quantity) {  
15        this.id = id;  
16        this.name = name;  
17        this.category = category;  
18        this.price = price;  
19        this.quantity = quantity;  
20    }  
21    // Constructor tanpa ID (Untuk Insert)  
22  
23    public Product(String name, String category, double price, int quantity) {  
24        this.name = name;  
25        this.category = category;  
26        this.price = price;  
27        this.quantity = quantity;  
28    }  
29    // Getter dan Setter  
30  
31    public int getId() {  
32        return id;  
33    }  
34  
35    public void setId(int id) {  
36        this.id = id;  
37    }  
38  
39    public String getName() {  
40        return name;  
41    }  
42  
43    public void setName(String name) {  
44        this.name = name;  
45    }  
46  
47    public String getCategory() {  
48        return category;  
49    }  
50 }
```

Figure 37 Product Servlet

```
50
51  [ public void setCategory(String category) {
52      [     this.category = category;
53      ]
54  }
55  [ public double getPrice() {
56      [     return price;
57      ]
58  }
59  [ public void setPrice(double price) {
60      [     this.price = price;
61      ]
62  }
63  [ public int getQuantity(){
64      [     return quantity;
65      ]
66  }
67  [ public void setQuantity(int quantity){
68      [     this.quantity = quantity;
69      ]
70  }
71  }
72
```

Figure 38 Product Servlet

```
1 package com.lab.dao;
2
3 import com.lab.model.Product;
4 import java.sql.Connection;
5 import java.sql.DriverManager;
6 import java.sql.PreparedStatement;
7 import java.sql.ResultSet;
8 import java.sql.SQLException;
9 import java.util.ArrayList;
10 import java.util.List;
11
12 public class ProductDAO {
13     private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
14     private String jdbcUsername = "root";
15     private String jdbcPassword = "admin";
16
17     protected Connection getConnection() {
18         Connection connection = null;
19         try {
20             Class.forName("com.mysql.cj.jdbc.Driver");
21             connection = DriverManager.getConnection(jdbcURL,
22                                                     jdbcUsername, jdbcPassword);
23         } catch (SQLException | ClassNotFoundException e) {
24             e.printStackTrace();
25         }
26         return connection;
27     }
28
29     public void insertProduct(Product product) {
30         String sql = "INSERT INTO products (name, category, price, quantity) VALUES( ?, ?, ?, ?)";
31
32         try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
33             pstmt.setString(1, product.getName());
34             pstmt.setString(2, product.getCategory());
35             pstmt.setDouble(3, product.getPrice());
36             pstmt.setInt(4, product.getQuantity());
37
38             pstmt.executeUpdate();
39         } catch (SQLException e) {
40             e.printStackTrace();
41         }
42     }
43
44     public Product selectUser(int id) {
45         Product product = null;
46         String sql = "SELECT id, name, category, price, quantity FROM products WHERE id = ? ";
47
48         try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
49             pstmt.setInt(1, id);
```

Figure 39 Product DAOServlet

```

49         pstmt.setInt(1, id);
50         ResultSet rs = pstmt.executeQuery();
51         if (rs.next()) {
52             String name = rs.getString("name");
53             String category = rs.getString("category");
54             Double price = rs.getDouble("price");
55             int quantity = rs.getInt("quantity");
56             product = new Product(id, name, category, price, quantity);
57         }
58     } catch (SQLException e) {
59         e.printStackTrace();
60     }
61     return product;
62 }
63 // READ: Ambil senarai semua pengguna (Untuk View)
64
65 public List<Product> selectAllProducts() {
66     List<Product> products = new ArrayList<>();
67     String sql = "SELECT * FROM products";
68     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
69         ResultSet rs = pstmt.executeQuery();
70         while (rs.next()) {
71             int id = rs.getInt("id");
72             String name = rs.getString("name");
73             String category = rs.getString("category");
74             Double price = rs.getDouble("price");
75             int quantity = rs.getInt("quantity");
76             products.add(new Product(id, name, category, price, quantity));
77         }
78     } catch (SQLException e) {
79         e.printStackTrace();
80     }
81     return products;
82 }
83 // UPDATE: Kemaskini maklumat pengguna
84
85 public boolean updateProduct(Product product) {
86     boolean rowUpdated = false;
87     String sql = "UPDATE products SET name = ?, category = ?, price = ?, quantity = ? WHERE id = ?";
88
89     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
90         pstmt.setString(1, product.getName());
91         pstmt.setString(2, product.getCategory());
92         pstmt.setDouble(3, product.getPrice());
93         pstmt.setInt(4, product.getQuantity());
94         pstmt.setInt(5, product.getId());
95         rowUpdated = pstmt.executeUpdate() > 0;
96     } catch (SQLException e) {
97         e.printStackTrace();
98     }
99     return rowUpdated;
100 }
101 // DELETE: Padam pengguna
102
103 public boolean deleteProduct(int id) {
104     boolean rowDeleted = false;
105     String sql = "DELETE FROM products WHERE id = ?";
106     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
107         pstmt.setInt(1, id);
108         rowDeleted = pstmt.executeUpdate() > 0;
109     } catch (SQLException e) {
110         e.printStackTrace();
111     }
112     return rowDeleted;
113 }
114 }
115

```

Figure 40 Product DAOServlet

Cs

```

97     } catch (SQLException e) {
98         e.printStackTrace();
99     }
100     return rowUpdated;
101 }
102 // DELETE: Padam pengguna
103
104 public boolean deleteProduct(int id) {
105     boolean rowDeleted = false;
106     String sql = "DELETE FROM products WHERE id = ?";
107     try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
108         pstmt.setInt(1, id);
109         rowDeleted = pstmt.executeUpdate() > 0;
110     } catch (SQLException e) {
111         e.printStackTrace();
112     }
113     return rowDeleted;
114 }
115

```

Figure 41 Product DAOServlet

```
...va ViewServlet.java x Product.java x ProductDAO.java x InsertProductServlet.java x ViewProductServlet.java x UpdateProductServlet.java x DeleteProductServlet.java x
Source History
1 import com.lab.dao.ProductDAO;
2 import java.io.IOException;
3 import java.io.PrintWriter;
4 import javax.servlet.ServletException;
5 import javax.servlet.http.HttpServlet;
6 import javax.servlet.http.HttpServletRequest;
7 import javax.servlet.http.HttpServletResponse;
8
9
10 public class DeleteProductServlet extends HttpServlet {
11     private ProductDAO productDAO;
12
13     public void init() {
14         productDAO = new ProductDAO();
15     }
16
17     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
18         throws ServletException, IOException {
19         response.setContentType("text/html;charset=UTF-8");
20         try (PrintWriter out = response.getWriter()) {
21             /* TODO output your page here. You may use following sample code. */
22             out.println("<!DOCTYPE html>");
23             out.println("<html>");
24             out.println("<head>");
25             out.println("<title>Servlet DeleteProductServlet</title>");
26             out.println("</head>");
27             out.println("<body>");
28             out.println("<h1>Servlet DeleteProductServlet at " + request.getContextPath() + "</h1>");
29             out.println("</body>");
30             out.println("</html>");
31         }
32     }
33
34     @Override
35     protected void doGet(HttpServletRequest request, HttpServletResponse response)
36         throws ServletException, IOException {
37         int id = Integer.parseInt(request.getParameter("id"));
38         productDAO.deleteProduct(id); // Panggil DAO
39         response.sendRedirect("ViewProductServlet");
40     }
41
42     @Override
43     protected void doPost(HttpServletRequest request, HttpServletResponse response)
44         throws ServletException, IOException {
45         processRequest(request, response);
46     }
47 }
```

```
47
48
49 @Override
50 public String getServletInfo() {
51     return "Short description";
52 } // </editor-fold>
53
54 }
```

Figure 42 DeleteProductServlet

```

1 import com.lab.dao.ProductDAO;
2 import com.lab.model.Product;
3 import java.io.IOException;
4 import java.io.PrintWriter;
5 import javax.servlet.ServletException;
6 import javax.servlet.http.HttpServlet;
7 import javax.servlet.http.HttpServletRequest;
8 import javax.servlet.http.HttpServletResponse;
9
10 public class UpdateProductServlet extends HttpServlet {
11
12     private ProductDAO productDAO;
13
14     public void init() {
15         productDAO = new ProductDAO();
16     }
17
18     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
19         throws ServletException, IOException {
20         response.setContentType("text/html;charset=UTF-8");
21         try (PrintWriter out = response.getWriter()) {
22             /* TODO output your page here. You may use following sample code. */
23             out.println("<!DOCTYPE html>");
24             out.println("<html>");
25             out.println("<head>");
26             out.println("<title>Servlet UpdateProductServlet</title>");
27             out.println("</head>");
28             out.println("<body>");
29             out.println("<h1>Servlet UpdateProductServlet at " + request.getContextPath() + "</h1>");
30             out.println("</body>");
31             out.println("</html>");
32         }
33     }
34
35     @Override
36     protected void doGet(HttpServletRequest request, HttpServletResponse response)
37         throws ServletException, IOException {
38         response.setContentType("text/html");
39         PrintWriter out = response.getWriter();
40         int id = Integer.parseInt(request.getParameter("id"));
41         Product existingProduct = productDAO.selectUser(id); // Panggil DAO
42         if (existingProduct != null) {
43             out.println("<h2>Update Product (Using DAO)</h2>");
44             out.println("<form action='UpdateProductServlet' method='POST'>");
45             out.println("<input type='hidden' name='id' value='" +
46                 existingProduct.getId() + "'>");
47             out.println("Name: <input type='text' name='name' value='" + existingProduct.getName() + "' required><br><br>");
48             out.println("Category: <input type='text' name='category' value='" + existingProduct.getCategory() + "' required><br><br>");
49             out.println("Category: <input type='text' name='category' value='" + existingProduct.getCategory() + "' required><br><br>");
50             out.println("Price: <input type='text' name='price' value='" + existingProduct.getPrice() + "' required><br><br>");
51             out.println("Quantity: <input type='text' name='quantity' value='" + existingProduct.getQuantity() + "' required><br><br>");
52             out.println("<input type='submit' value='Update Product'>");
53             out.println("</form>");
54         }
55     }
56
57     @Override
58     protected void doPost(HttpServletRequest request, HttpServletResponse response)
59         throws ServletException, IOException {
60         int id = Integer.parseInt(request.getParameter("id"));
61         String name = request.getParameter("name");
62         String category = request.getParameter("category");
63         Double price = Double.parseDouble(request.getParameter("price"));
64         int quantity = Integer.parseInt(request.getParameter("quantity"));
65         Product product = new Product(id, name, category, price, quantity);
66         productDAO.updateProduct(product); // Panggil DAO, bukan JDBC
67         response.sendRedirect("ViewProductServlet");
68     }
69
70     @Override
71     public String getServletInfo() {
72         return "Short description";
73     }
74 }

```

Figure 43 UpdateProductServlet

```
...va ViewServlet.java x Product.java x ProductDAO.java x InsertProductServlet.java x ViewProductServlet.java x UpdateProdu

Source History

1 package com.lab.controller;
2 import com.lab.dao.ProductDAO;
3 import com.lab.model.Product;
4 import java.io.IOException;
5 import java.io.PrintWriter;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10
11 public class InsertProductServlet extends HttpServlet {
12
13     private ProductDAO productDAO;
14
15     public void init() {
16         productDAO = new ProductDAO();
17     }
18
19     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
20         throws ServletException, IOException {
21         response.setContentType("text/html;charset=UTF-8");
22         try (PrintWriter out = response.getWriter()) {
23             /* TODO output your page here. You may use following sample code. */
24             out.println("<!DOCTYPE html>");
25             out.println("<html>");
26             out.println("<head>");
27             out.println("<title>Servlet InsertProductServlet</title>");
28             out.println("</head>");
29             out.println("<body>");
30             out.println("<h1>Servlet InsertProductServlet at " + request.getContextPath() + "</h1>");
31             out.println("</body>");
32             out.println("</html>");
33         }
34     }
35
36     @Override
37     protected void doGet(HttpServletRequest request, HttpServletResponse response)
38         throws ServletException, IOException {
39         processRequest(request, response);
40     }
41
42     @Override
43     protected void doPost(HttpServletRequest request, HttpServletResponse response)
44         throws ServletException, IOException {
45         String name = request.getParameter("name");
46         String category = request.getParameter("category");
47         Double price = Double.parseDouble(request.getParameter("price"));
48         int quantity = Integer.parseInt(request.getParameter("quantity"));
49         Product newProduct = new Product(name, category, price, quantity);
50
51         productDAO.insertProduct(newProduct); // Panggil DAO, bukan JDBC
52         response.sendRedirect("ViewProductServlet");
53     }
54
55     @Override
56     public String getServletInfo() {
57         return "Short description";
58     }
59 }
60
```

Figure 44 InsertProductServlet

```

1 package com.lab.controller;
2 import com.lab.dao.ProductDAO;
3 import com.lab.model.Product;
4 import java.io.IOException;
5 import java.io.PrintWriter;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.HttpServlet;
8 import javax.servlet.http.HttpServletRequest;
9 import javax.servlet.http.HttpServletResponse;
10 import java.util.List;
11
12 public class ViewProductServlet extends HttpServlet {
13
14     private ProductDAO productDAO;
15
16     public void init() {
17         productDAO = new ProductDAO();
18     }
19
20     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22         response.setContentType("text/html;charset=UTF-8");
23         try (PrintWriter out = response.getWriter()) {
24             /* TODO output your page here. You may use following sample code. */
25             out.println("<!DOCTYPE html>");
26             out.println("<html>");
27             out.println("<head>");
28             out.println("<title>Servlet ViewProductServlet</title>");
29             out.println("</head>");
30             out.println("<body>");
31             out.println("<h1>Servlet ViewProductServlet at " + request.getContextPath() + "</h1>");
32             out.println("</body>");
33             out.println("</html>");
34         }
35     }
36
37     @Override
38     protected void doGet(HttpServletRequest request, HttpServletResponse response)
39         throws ServletException, IOException {
40         response.setContentType("text/html");
41         PrintWriter out = response.getWriter();
42
43         List<Product> listProduct = productDAO.selectAllProducts(); // Panggil DAO
44
45         out.println("<h2>Product List</h2>");
46         out.println("<table border='1'><tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Quantity</th><th>Actions</th></tr>");
47
48         for (Product product : listProduct) {
49
50             out.println("<tr>");
51             out.println("<td>" + product.getId() + "</td>");
52             out.println("<td>" + product.getName() + "</td>");
53             out.println("<td>" + product.getCategory() + "</td>");
54             out.println("<td>" + product.getPrice() + "</td>");
55             out.println("<td>" + product.getQuantity() + "</td>");
56             out.println("<td><a href='UpdateProductServlet?id=" + product.getId() + "'>Edit</a> | ");
57             out.println("<a href='DeleteProductServlet?id=" + product.getId() + "'>Delete</a></td>");
58             out.println("</tr>");
59
60         }
61         out.println("</table>");
62         out.println("<br><a href='add_product.html'>Add New Product</a>");
63     }
64
65     @Override
66     protected void doPost(HttpServletRequest request, HttpServletResponse response)
67         throws ServletException, IOException {
68         processRequest(request, response);
69     }
70
71     @Override
72     public String getServletInfo() {
73         return "Short description";
74     } // </editor-fold>
75
76 }

```

Figure 45 ViewProductServlet

```

1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template
5 -->
6 <html>
7 <head>
8 <title>TODO supply a title</title>
9 <meta charset="UTF-8">
10 <meta name="viewport" content="width=device-width, initial-scale=1.0">
11 </head>
12 <body>
13 <h2>ADD Product</h2>
14 <form action="InsertProductServlet" method="POST">
15 <label for="name">Name:</label>
16 <input type="text" id="name" name="name"><br><br>
17 <label for="category">Category:</label>
18 <input type="text" id="category" name="category"><br><br>
19 <label for="price">Price:</label>
20 <input type="text" id="price" name="price"><br><br>
21 <label for="quantity">Quantity:</label>
22 <input type="text" id="quantity" name="quantity"><br><br>
23 <input type="submit" value="Add Product">
24 </form>
25 </body>
26 </html>
27

```

Figure 46 add_product.html

OUTPUT:

ADD USER

Username:

Password:

Roles:

[Go to Add Product Page](#)

Add product:

ADD Product

Name:

Category:

Price:

Quantity:

Before update:

Product List

< / tr >

ID	Name	Category	Price	Quantity	Actions
3	bread	food	2.5	10	Edit Delete

[Add New Product](#)

Update page:

Update Product (Using DAO)

Name:

Category:

Price:

Quantity:

After update:

Product List

< / tr >

ID	Name	Category	Price	Quantity	Actions
4	bread	food	2.5	100	Edit Delete

[Add New Product](#)

Before delete:

Product List

< / tr >

ID	Name	Category	Price	Quantity	Actions
4	bread	food	2.5	100	Edit Delete

[Add New Product](#)

After delete:

Product List

< / tr >

ID	Name	Category	Price	Quantity	Actions
----	------	----------	-------	----------	---------

[Add New Product](#)